

What Held-Out Accuracy Predicts About Deploying Intrusion Detection in O-RAN

ARSHAD ALI¹ AND ADEEL AHMAD¹

¹Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia

ABSTRACT Papers on intrusion detection for 5G and O-RAN often report more than 99% accuracy on a random split of one corpus. We asked what such a number says about deployment. Our study uses two public corpora (NetsLab-5GORAN-IDD and 5G-NIDD), six architectures, two baselines that ignore their input, and increasingly strict test protocols. On the radio layer, a random split adds 0.13 to macro- F_1 (95% interval -0.01 to 0.27, significant only for two sequence models), and a nearest-neighbor lookup gains 0.15, consistent with the recognition of capture sessions. False positive rates on held-out benign sessions range from 0.00 to 0.98. In 5G-NIDD, 59% of the benign flows are copies of flood records labeled as attacks, and their addresses mark them as the flood of a second attacker. These copies cap balanced accuracy at 0.766. The published 99.9% accuracy rests on two fields that encode a record's position in its file, and without them it drops to 76.9%. On flows with consistent labels, transferred detectors reach a balanced accuracy of only 0.61 to 0.78. Re-extracting the target with the source's exporter does not close this gap, and a third corpus behaves the same way. At an attack prevalence of 0.002, precision never exceeds 0.063 while recall stays above 10%, and no detector passes our deployability test. Latency is the exception. In a FlexRIC near-real-time RIC with an emulated E2 node, a decision and its control message take at most 5.01 ms at the 99th percentile. We close with reporting requirements.

INDEX TERMS Dataset shift, evaluation methodology, false positive rate, intrusion detection, label quality, O-RAN, RAN intelligent controller, xApp.

I. INTRODUCTION

Open Radio Access Network (O-RAN) splits the base station into radio, distributed, and central units joined by open interfaces, and it adds a RAN Intelligent Controller (RIC) that runs third-party control applications [1]–[3]. The near-real-time RIC sits close to the cells it controls [1], [4], so it sees the telemetry of every device those cells serve. That makes it an obvious place to look for attacks. Compromised devices, many of them IoT devices, are a well-documented source of volumetric attacks [5], [6], and a detection application (xApp) inside the RIC control loop could catch them close to where they enter the network. Platforms for building xApps exist [7], [8], and detectors that run in or beside the O-RAN control loop have already been shown to work [9], [10].

Most of these detectors are tested on a random held-out split of a single dataset, where accuracy or F_1 above 99% is common. For 5G-NIDD, one of our two corpora, the dataset authors report binary accuracies of 99.85% to 99.95% for four of five classifiers [11], and later work reports a weighted F_1 of up to 99.95% [12]. Such a split says little about deployment. Three properties decide whether a detector can be deployed, and the split measures none of them. The first is generalization to a site that contributed no training data, and intrusion

detectors are known to transfer poorly between independently collected datasets [13], [14]. The second is alert burden: a false positive rate that looks small on a balanced benchmark becomes a stream of alerts once it meets production flow rates and a realistic attack base rate [15], [16]. The third is latency, because the O-RAN E2 specification requires support for control loops between 10 ms and 1 s [17]. A benchmark score measures how well a detector separates the records of one corpus. Deployment asks a different question, and we treat a benchmark result as evidence about deployment only when an experiment supports that reading.

None of these concerns is new. Axelsson made the base-rate argument [15] and Sommer and Paxson the closed-world critique [16]. Arp et al. catalogued the common pitfalls of machine learning in security [18], and Pendlebury et al. formalized temporal and spatial bias [19]. For the two corpora used here, Abraheem and Edhirig measured transfer [20], and Obiwewwi et al. timed inference inside a real near-real-time RIC [21]. We offer neither a new detector nor the finding that transfer hurts. What we add is a single controlled evaluation of one detector family in which these effects are measured together, with input-blind baselines on every corpus, so that their sizes can be compared and the claims that rest on them

can be checked. Our contributions are these:

- A ladder of evaluation protocols applied to one pipeline, six architectures, and two input-blind baselines. A model-free nearest-neighbor probe and two sequence models trace the random-split gain to the recognition of capture sessions (Section VI-A). Once category coverage is held fixed, the radio corpus supports no estimate of temporal drift, and its false positive rate depends mainly on which benign session is tested (Section VI-B).
- An audit of the target corpus. Half of the 5G-NIDD flows lie on records that occur under both labels, the addresses of the benign copies identify them as the flood of a second attacker, and a published 99.9% result depends on record-position fields. We then move the published pipeline down the same ladder (Section VI-C).
- A breakdown of the cross-corpus gap into prior shift, label conflicts, and genuine transfer failure. It uses a prevalence-invariant metric, a difference-in-differences test against the majority baseline, references trained inside the target, and evaluation with and without the conflicting records. We also re-extract the target from its raw captures with the source's exporter, and we add a third corpus from another site with another exporter (Sections VI-D, VI-E, and VI-F).
- Operational precision and alert volume from pooled confusion counts, in explicit units and with cluster-bootstrap intervals. Averaging precision over folds instead inflates the apparent difference between architectures by an order of magnitude (Section VI-G).
- An account of which calibration methods can change the reachable operating points. None of them helps here, and label-free estimates of the target prior fail once the class-conditional distributions shift (Section VI-H).
- Decision latency on one host for three implementations, one of them a compiled runtime, and the full decision loop of the detector running as an xApp in a FlexRIC near-real-time RIC with an emulated E2 node (Section VI-I). Finally, a deployability test that combines generalization, alert burden, and latency, an analysis of how its verdict depends on its bounds, and a set of reporting requirements (Sections VII and X).

Two limits of scope apply throughout. We measured latency on one host, once without a RIC and once through FlexRIC with an emulated E2 node, so no radio or operator network took part. The three corpora also come from three different flow exporters, which means that every cross-corpus gap mixes a change of site with a change of tooling. For $\mathcal{D}_A$ and $\mathcal{D}_B$ we remove this confound by re-extracting $\mathcal{D}_B$ with the exporter of $\mathcal{D}_A$ (Section VI-E). Table 1 lists the notation.

II. BACKGROUND AND RELATED WORK

Four strands of work matter here: the RIC as a place to run detectors, the security of O-RAN, the datasets and models of network intrusion detection, and the literature on evaluation practice and dataset shift that our methods borrow from.

TABLE 1. Notation and abbreviations.

Symbol	Meaning
$\mathcal{D}_A, \mathcal{D}_B$	source corpus (NetsLab-5GORAN-IDD); target corpus (5G-NIDD)
$\mathcal{D}_C$	third corpus, a second target (5G core datasets of [22])
τ	decision threshold on the attack score $\hat{p}$
$R(\tau), \text{FPR}(\tau)$	recall and false positive rate at τ
π	operational attack prevalence (declared, not measured)
π_{tr}	attack prevalence of the training data
λ_b	benign arrival rate (flows or windows per hour)
$A(\tau)$	false alerts per hour at τ
$\text{PPV}(\tau)$	operational precision at prevalence π
Δ_{BA}, Δ_{F1}	source minus target balanced accuracy, macro- F_1
$\bar{W}_1$	mean first-order Wasserstein distance over features
$L, q_p(L)$	decision latency and its p -th quantile
B, ε	latency budget and tolerated violation rate
$\delta, A_{\max}, r_{\min}, \rho$	bounds of the deployability test, Eq. (6)
δ_m	deadline margin in Algorithm 1 (design, not evaluated)
BA	balanced accuracy, the mean of TPR and TNR
LR, DT, RF	logistic regression, decision tree, random forest
XGB, HGB, MLP	XGBoost, histogram gradient boosting, multilayer perceptron
IF, AE, GRU	isolation forest, autoencoder, gated recurrent unit
KPM, RC	E2 service models for measurements and RAN control
UE	user equipment

A. THE NEAR-REAL-TIME RIC AS A DETECTION PLATFORM

An xApp talks to E2 nodes through service models. E2SM-KPM delivers periodic key performance measurements, and E2SM-RC exposes control actions [1]. Building experimental xApps became practical with platforms such as OpenRAN Gym [23], SCOPE [24], CoLo-RAN [7], and FlexRIC [8], together with open radio stacks [25], [26]. One constraint shapes everything that follows. E2 carries aggregated measurements, not packets, so a flow-level detector needs a data path of its own (Section VII-A).

B. SECURITY OF OPEN RAN

The RIC and its third-party xApps are themselves a new attack surface, according to the O-RAN Alliance threat model [27] and several independent analyses [28], [29], and security mechanisms for the O-RAN interfaces have since been built and evaluated [30]. On the detection side, 5G-Spector offers layer-3 attack detection as an O-RAN compliant service [9], Det-RAN detects attacks from cross-layer data in real time [10], and Fard et al. fuse radio telemetry and flow records on the same source corpus we use [31]. All of them build or test a detector inside one deployment. Our question is what evidence would show that such a detector still works in another.

C. MACHINE LEARNING FOR NETWORK INTRUSION DETECTION

Public intrusion corpora are plentiful for IoT traffic [32]–[36] and scarce for cellular networks. 5G-NIDD was captured on an operational 5G test network [37], [38], while NetsLab-5GORAN-IDD pairs flow records with radio telemetry from an O-RAN testbed [39]. On 5G-NIDD, the dataset authors report binary accuracies of 99.85% to 99.95% for four of five classifiers on a random 70/30 split [11], and later work reaches a weighted F_1 of up to 99.95% [12]. Standardized

NetFlow feature sets made cross-corpus experiments possible [40], [41]. The corpora can be wrong, however. Engelen et al. documented labeling and flow-construction faults in CIC-IDS2017 [42], and Flood et al. found design flaws across NIDS benchmarks that let a classifier score well without learning the attack [43].

D. EVALUATION PRACTICE AND DATASET SHIFT

Reported results tend to overstate what a detector can do, and the causes are well known: closed-world evaluation [16], neglect of the base rate [15], sampling bias and data leakage [18], [44], and temporal or spatial bias in the split [19]. Apruzzese et al. trace the gap between reported and operational effectiveness across security domains [45], Kus et al. see the same gap in industrial intrusion detection once detectors meet data they were not tuned on [46], and cross-dataset generalization has repeatedly proved poor [13], [14], [47]. Closest to our work, Abraheem and Edhirig transfer XGBoost between the same two corpora with 15 harmonized features and report zero-shot balanced accuracies per attack family of 0.50 to 0.56 from NetsLab-5GORAN-IDD to 5G-NIDD and 0.52 to 0.83 in the other direction [20]. Our transfer results match theirs in direction and size. We add the protocol ladder with input-blind baselines, references trained inside the target, the published pipeline under the same ladder, an audit of the target labels, and the operational and latency criteria. From the dataset-shift literature we take label-shift estimation from unlabeled data [48], [49], calibration and its behavior under shift [50]–[52], shift detection [53], and covariance alignment for domain adaptation [54]. Classifiers compared under repeated resampling are tested as Nadeau and Bengio recommend [55].

III. DEPLOYMENT CRITERIA AND THREAT MODEL

We state the system and threat model, define the three quantities we measure, and combine them into a deployability test.

A. SYSTEM AND THREAT MODEL

A near-real-time RIC serves a set of E2 nodes (Fig. 1), and user devices, IoT devices among them, attach to those nodes. The adversary controls some of these devices and uses them for flooding, scanning, brute-force login, or web attacks, the categories present in our corpora. It does not control the RIC, the E2 interface, or the xApp. The defender's xApp reads either radio telemetry windows delivered through E2SM-KPM or flow records from an exporter placed with the user plane, and it can act through E2SM-RC. That exporter has to live outside the RIC, since E2 carries no packets. Section VII-A returns to its cost and placement.

B. CRITERION 1: GENERALIZATION ACROSS DEPLOYMENTS

Let $\mathcal{D}_A$ and $\mathcal{D}_B$ be two independently collected corpora in a shared feature space. We describe a model f_θ trained on $\mathcal{D}_A$ by how far its score on $\mathcal{D}_B$ falls below its held-out score on $\mathcal{D}_A$. The two corpora differ in attack prevalence (Table 2),

which complicates the choice of metric. A change in macro- F_1 blends a change in the classifier with a change in class balance, and even a classifier that ignores its input scores differently when the prevalence moves. We therefore use the difference in balanced accuracy, which prevalence does not affect,

$$\Delta_{BA} = BA(f_\theta; \mathcal{D}_A^{\text{test}}) - BA(f_\theta; \mathcal{D}_B), \quad (1)$$

and we also give the macro- F_1 gap Δ_{F_1} and its difference from the gap of the majority-class baseline. Covariate shift is measured one feature at a time,

$$\bar{W}_1 = \frac{1}{|\mathcal{F}|} \sum_{j \in \mathcal{F}} w_1(Q_j(P_A^{(j)}), Q_j(P_B^{(j)})), \quad (2)$$

where Q_j is the quantile transform of feature j , fitted on source training data only. Had each corpus been normalized by its own quantiles, both marginals would become uniform and every term would vanish.

C. CRITERION 2: ALERT BURDEN UNDER A REALISTIC BASE RATE

Let λ_b be the benign arrival rate in the detector's own unit, which is flow records per hour for a flow-level detector and telemetry windows per hour for a window-level one. A benign UE produces 225 windows of 16 s per hour. False alerts then arrive at

$$A(\tau) = \lambda_b \text{FPR}(\tau), \quad (3)$$

and, for an operational attack prevalence π , the fraction of alerts that correspond to attacks is

$$\text{PPV}(\tau) = \frac{\pi R(\tau)}{\pi R(\tau) + (1 - \pi) \text{FPR}(\tau)}. \quad (4)$$

The corpora cannot give us π or λ_b , so we declare both and vary them over a range. PPV depends only on π , and A grows linearly with λ_b . For that reason we also report false alerts per 10^6 benign flows, a number that does not depend on λ_b .

D. CRITERION 3: LATENCY AGAINST THE CONTROL BUDGET

The decision latency of an xApp can be decomposed as

$$L = t_{\text{ind}} + t_{\text{feat}} + t_{\text{inf}} + t_{\text{act}} + t_q, \quad (5)$$

where the terms are the delivery of the indication to the xApp, feature construction, inference, the control action up to its acknowledgment, and queueing inside the xApp. We time feature construction and inference alone on one host, and all five terms in a loop through a near-real-time RIC with an emulated E2 node (Section V). What matters for conformance is the tail. For a budget B and a tolerated violation rate ε , we require $q_{1-\varepsilon}(L) \leq B$. The O-RAN E2 specification allows near-real-time latency requirements anywhere from 10 ms to 1 s [17], so instead of a verdict for one chosen value we report the smallest budget that is met.

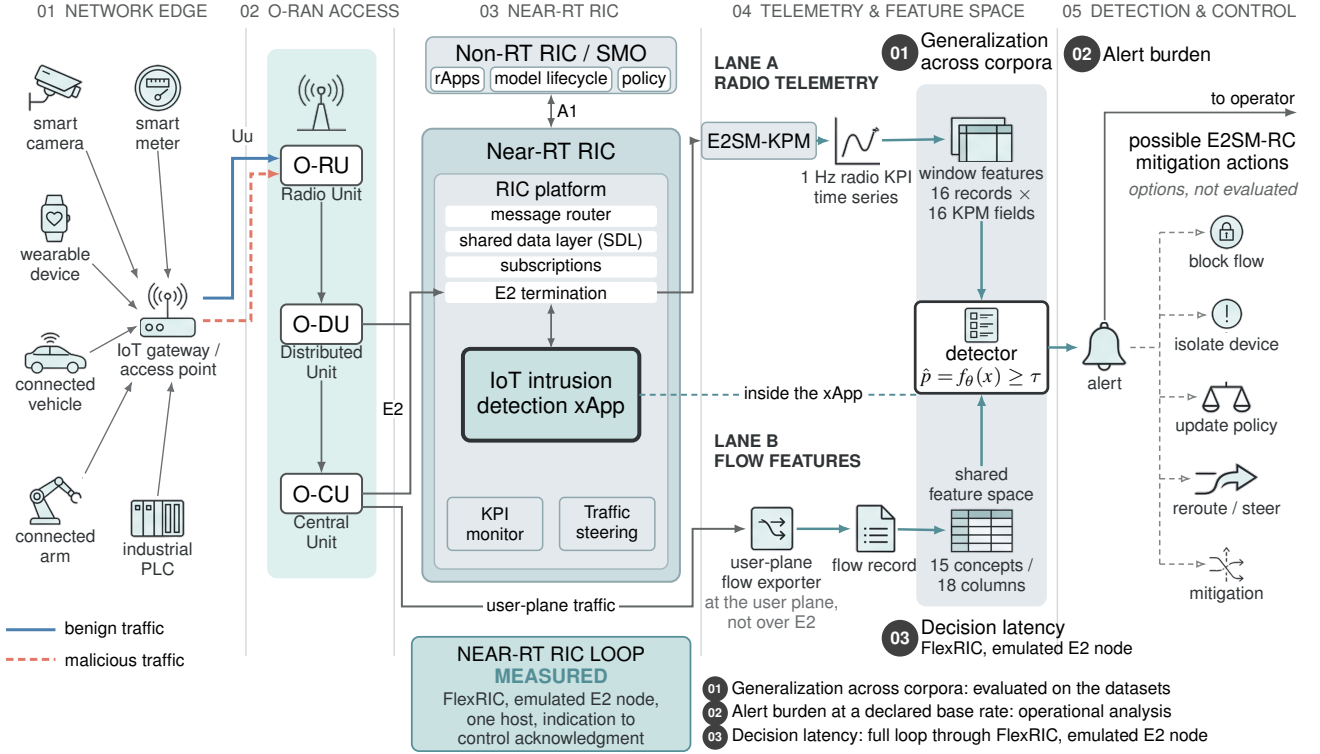


FIGURE 1. Reference O-RAN edge deployment and the detection loop. All device traffic reaches the radio unit over Uu. The detection xApp receives radio telemetry through the E2 termination (lane A) and flow records from an exporter at the user plane, mapped onto the shared feature space (lane B). Column 04 expands the xApp's own feature construction and inference, with one detector per layer. Numbered markers show what this paper measures. Generalization across corpora (01) and alert burden (02) are evaluated on the corpora. Decision latency (03) is timed on one host, from the indication to the acknowledgment of the control message, through a FlexRIC near-real-time RIC with an emulated E2 node. The mitigation actions on the right are options E2SM-RC offers and are not evaluated.

E. A DEPLOYABILITY TEST

The three criteria are combined into a test that a detector either passes or fails:

$$\begin{aligned} \mathcal{P}(f_\theta) = & \mathbf{1}[\bar{\Delta}_{BA} \leq \delta] \cdot \mathbf{1}[\bar{q}_{0.99}(L) \leq B] \\ & \cdot \mathbf{1}[\exists \tau : R(\tau) \geq r_{\min}, A(\tau) \leq A_{\max}, \\ & \text{PPV}(\tau) \geq \rho], \end{aligned} \quad (6)$$

where $\bar{\Delta}_{BA}$ and $\bar{q}_{0.99}$ are upper 95% confidence bounds, so a pass cannot hinge on a lucky point estimate, and every alert condition is evaluated on the target corpus. The minimum recall $r_{\min}$ keeps a detector that never raises an alert from passing the alert term. As operating threshold we take the most sensitive admissible one,

$$\begin{aligned} \tau^* = \arg \max_{\tau} R(\tau) \quad \text{s.t.} \quad & R(\tau) \geq r_{\min}, A(\tau) \leq A_{\max}, \\ & \text{PPV}(\tau) \geq \rho. \end{aligned} \quad (7)$$

We intend this test as a reporting instrument, not as an O-RAN requirement. Its bounds stand for operator requirements. Section VII-B gives the values we used and shows how the verdict responds when they change.

IV. METHODOLOGY

A. CORPORA

$\mathcal{D}_A$ is NetsLab-5GORAN-IDD [39], recorded on a physical OpenAirInterface O-RAN testbed. It holds benign traffic and five attack categories: denial of service (DoS), distributed denial of service (DDoS), probing, brute force, and web attacks. The corpus has two layers, and the published files do not let us join them record by record. One is flow records exported by Zeek at the central unit. The other is radio telemetry sampled at 1 Hz at the distributed unit. We use both. For the radio layer we form windows of 16 consecutive records of one UE and summarize each by the mean and standard deviation of 16 KPM fields.

Gaps of more than five minutes in the timestamps split the radio layer into 30 capture sessions. Fard et al. describe the same corpus as 42 experiment runs [31], and a gap threshold of 90 to 105 s does recover exactly 42 segments, each with a single attack category. Every one of our sessions is a union of complete segments. The five-minute rule merges runs recorded back to back (10 of the 30 sessions contain two or three attack subcategories), which keeps such runs on the same side of any split. All records of a session carry the same category label. A radio label is therefore a session label, and every window of an attack session is an attack window. The sessions follow the capture schedule (Section VI-B): nine benign sessions on days 0 to 6, the attack categories in blocks

TABLE 2. Datasets. $\mathcal{D}_A$ supplies training and in-distribution testing on both layers. $\mathcal{D}_B$ and $\mathcal{D}_C$ are transfer targets; each also trains its own in-target references, and $\mathcal{D}_B$ trains the reverse direction.

	$\mathcal{D}_A$ network	$\mathcal{D}_A$ radio	$\mathcal{D}_B$	$\mathcal{D}_C$
Unit	flow	16 s UE window	flow	flow
Source	Zeek	KPM, 1 Hz	Argus	NFStream
Samples	1 640 182	2 808	1 215 889	39 425
Features	18	32	18	18
Attack prevalence	94.63%	76.46%	60.71%	2.13%
Attack categories	5	5	2	3
Groups	318 hosts	30 runs	20 files	3 files
Role	train, test	train, test	target	target

on days 47 to 53, and one last benign session.

$\mathcal{D}_B$ is 5G-NIDD [37], [38], captured on a 5G test network at another institution and exported by Argus, with two attack categories (DoS and probing). Its combined files have no addresses, ports, or timestamps. They do keep the Argus record counter and file offset, and because the offset restarts at every file boundary we can recover 20 capture files, two passes over the same ten attack sessions, one per base station. The row index of `Combined.csv` also restarts at the second pass. Each attack file holds benign traffic plus one attack type. A second release of the corpus keeps every Argus field, addresses included, and the raw captures are public as well (Sections VI-C and VI-E).

A third corpus, $\mathcal{D}_C$, gives us a second transfer target at a third site. It contains the flow records of a 5G core testbed (Open5GS in Docker) published by Nugraha et al. [22]. Only three of its files hold flow records, each with one attack (a SYN flood, an ICMP flood, and a PFCP session-deletion attack), for 39,425 flows in total, 839 of them attacks. Its benign flows mix user traffic with the signaling of the core network. NFStream exported $\mathcal{D}_C$, and it counts link-layer bytes plus, for flows inside the GTP-U user-plane tunnel, 44 bytes of encapsulation per packet. We strip both overheads so that the shared columns hold the IP bytes of the inner packet, as they do for $\mathcal{D}_A$. Table 2 summarizes the three datasets.

B. SHARED FEATURE SPACE

Zeek and Argus share no column names, so we define the flow feature space by what each field means (Table 3). It has 15 concepts in 18 columns: seven counts that both exporters report, a one-hot protocol with a fixed list of levels, and seven derived quantities computed the same way for both corpora. One mapping is easy to get wrong. Zeek's `src_bytes` counts application payload, which is zero for most floods and scans, whereas Argus's `SrcBytes` counts whole packets. The true counterpart of `SrcBytes` is Zeek's `src_ip_bytes`, at 40 and 42 bytes per packet, respectively. Both corpora get the same lower bound on duration and the same $\log(1 + x)$ transform for unbounded quantities, and scaling is fitted on source training data alone. No address, port, or timestamp is ever a feature.

Before computing any result with it, we also fixed an exporter-robust subset of nine columns: the protocol and the

TABLE 3. Shared flow feature space, generated from `configs/features/shared_space.yaml`. The exporter-robust subset is the protocol one-hot, the three mean packet sizes, and the two direction ratios.

Concept	$\mathcal{D}_A$ (Zeek)	$\mathcal{D}_B$ (Argus)	Role
duration	duration	Dur	base
src_bytes	src_ip_bytes	SrcBytes	base
dst_bytes	dst_ip_bytes	DstBytes	base
src_pkts	src_pkts	SrcPkts	base
dst_pkts	dst_pkts	DstPkts	base
tot_pkts	src_pkts + dst_pkts	TotPkts	base
tot_bytes	src_ip_bytes + dst_ip_bytes	TotBytes	base
proto	proto	Proto	one-hot (4)
mean_pkt_size	tot_bytes / max(tot_pkts, 1)		derived
src_mean_pkt_size	src_bytes / max(src_pkts, 1)		derived
dst_mean_pkt_size	dst_bytes / max(dst_pkts, 1)		derived
bytes_per_s	tot_bytes / max(duration, duration_floor_s)		derived
pkts_per_s	tot_pkts / max(duration, duration_floor_s)		derived
src_byte_ratio	src_bytes / max(tot_bytes, 1)		derived
src_pkt_ratio	src_pkts / max(tot_pkts, 1)		derived

five per-packet means and direction ratios. The two exporters end flows at different points, which moves duration, rates, and totals. Per-packet means and direction ratios move less, although they are not invariant. Splitting a flow changes them whenever its packets differ (handshake and teardown packets landing in different records, for example), and the exporters count the same packet as 40 and 42 bytes, an offset of about 5% for minimum-size packets.

C. DETECTORS

Every protocol is run on six supervised architectures and two trivial baselines. The supervised models are LR, a DT of depth 12, an RF [56], XGB [57], the HGB implementation of scikit-learn [58], and an MLP. The baselines are a majority-class predictor and a stratified sampler of the training prior. They ignore their input, so any change in their score between two protocols comes from class composition alone. All supervised models use balanced class weights (sample weights for the MLP, `scale_pos_weight` for XGB). We fixed the hyperparameters and did not tune them (Table 4). Tuning on held-out source data would reward source-specific structure, which is exactly what the transfer experiments measure, and tuning on the target would leak target information into training.

Three more model families answer narrower questions. Two sequence models, a GRU [59] and a one-dimensional convolutional network, read the raw 16-step KPM sequences of the same radio windows. Two unsupervised detectors, an isolation forest [60] and an autoencoder, learn from benign source flows only, and their alarm threshold sits at the 99th percentile of benign source scores. Finally, a CORAL variant [54] re-colors the source flows to the covariance of the unlabeled target before training. It is the one case in which a detector sees target data.

D. EVALUATION PROTOCOLS

Splits are group-disjoint wherever a group key exists, and an automated check rejects any split that puts a group on both sides. On the radio layer the group key is the capture session.

TABLE 4. Hyperparameters, identical across protocols and seeds.

Model	Settings
Majority, Strat.	training majority; samples of the training prior
LR	max_iter 2000, balanced weights, standardized
DT	depth 12, balanced weights
RF	200 trees, min. leaf 2, balanced subsample weights
XGB	300 trees, depth 6, $\eta = 0.1$, subsample 0.8, hist
HGB	300 iterations, depth 6, $\eta = 0.1$, balanced weights
MLP	128–64, Adam, early stopping (patience 10), balanced sample weights, standardized
GRU	hidden 32; BCE with positive weight n_-/n_+
1D-CNN	two convolutions (32, kernel 3), average pooling
IF	200 trees; benign-only training
AE	12–6–12 MLP, reconstruction error; benign-only
CORAL	per-domain z-score, identity-regularized covariances

We compare three protocols there: a random split of windows, a run-disjoint split, and a category-stratified run-disjoint split that keeps sessions of every attack category on both sides, so that no test category is missing from training. Two further designs probe the effect of time (Section VI-B). One holds out the latest or the earliest session of every category and compares the outcome with draws that hold out a random session per category. The other holds out each benign session on its own. On the flow layer the group key is the source address (318 groups). For each of 20 split seeds we draw 300,000 flows from $\mathcal{D}_A$, divide them by source address, and evaluate every model on the whole of $\mathcal{D}_B$. The reverse direction trains on a random split of $\mathcal{D}_B$ and tests on $\mathcal{D}_A$. Inside $\mathcal{D}_B$, the in-target references either hold out four of the 20 capture files or train on one base station's pass and test on the other's (Section VI-C). Apart from these references, CORAL, the domain classifier, and the prior estimates of Section VI-H (all marked as such), no transform, threshold, or model choice touches $\mathcal{D}_B$. Every access to it is logged.

E. STATISTICAL ANALYSIS

We estimate uncertainty over 20 split seeds. The splits resample one population of 30 sessions or 318 source addresses, so their training and test sets overlap, and a paired t -test over split means understates the variance. Our intervals are therefore Nadeau–Bengio corrected [55]: the variance is scaled by $1/J + n_{\text{test}}/n_{\text{train}}$ for J splits. Holm correction is applied within each family of tests, where a family is one metric and one comparison (a protocol pair or a transfer direction) across the six supervised architectures. The two sequence models form a family of their own. Operating points come from confusion counts pooled over folds, for reasons given in Section VI-G. A pooled count includes a sample once for every seed in which it is tested, and samples in the same cluster are correlated. A binomial interval would be far too narrow. We use a cluster bootstrap with 2,000 replicates instead, resampling with replacement both the split seeds and the clusters (radio sessions, $\mathcal{D}_A$ source addresses, or $\mathcal{D}_B$ capture files).

Algorithm 1 Budget-aware decision loop of a detection xApp

Require: indication stream $\mathcal{S}$, model f_θ , threshold τ , budget B , margin δ_m

Ensure: E2 control actions; violation count v ; latency record

```

1:  $\mathcal{W} \leftarrow$  empty per-UE window map;  $v \leftarrow 0$ 
2: for all indication  $m$  arriving on  $\mathcal{S}$  do
3:    $t_0 \leftarrow \text{Now}()$ ;  $u \leftarrow \text{UEOf}(m)$ 
4:    $\mathcal{W}[u] \leftarrow \text{UpdateWindow}(\mathcal{W}[u], m)$ 
5:    $x \leftarrow \text{Features}(\mathcal{W}[u])$ 
6:    $\hat{p} \leftarrow f_\theta(x)$ 
7:   if  $\hat{p} \geq \tau$  then
8:     SendE2(EncodeControl( $u, \hat{p}$ ))  $\triangleright$  even if late
9:   end if
10:   $\ell \leftarrow \text{Now}() - t_0$ 
11:  if  $\ell > B - \delta_m$  then
12:     $v \leftarrow v + 1$   $\triangleright$  deadline violation, recorded
13:  end if
14:  RecordLatency( $\ell$ )  $\triangleright$  every indication
15: end for
```

F. EXPERIMENTAL PIPELINE AND INFERENCE LOOP

Fig. 2 shows the experimental pipeline, and Algorithm 1 the decision loop of a detection xApp. A decision that misses the budget is still acted on and logged as a violation, because a late mitigation is usually better than none. An operator who would rather drop late actions can move the transmission step inside the deadline check. Our FlexRIC xApp (Section V) runs this loop with two changes made for timing. It scores held-out test windows in place of windows built from the synthetic KPM values, and it sends a control message after every report, whatever the score, so that t_{act} is measured each time.

V. EXPERIMENTAL SETUP

All measurements ran on one workstation: a 13th Gen Intel Core i9-13900H processor (14 cores, 20 threads), 47.6 GB of memory, and Windows 11 Pro build 26120, with Python 3.11.9, scikit-learn 1.8.0, XGBoost 3.2.0, NumPy 2.3.5, and ONNX Runtime 1.30.0. The sequence models were trained with PyTorch 2.14 on the CPU in a separate environment. The full package list ships with the results.

For the latency loop we built FlexRIC [8] v2.0.0 in WSL2 (Ubuntu, Linux kernel 6.18) on the same machine and connected its near-real-time RIC over SCTP to the emulated gNB agent that comes with FlexRIC. Our xApp is written in C. It subscribes to E2SM-KPM reports with 16 measurement items per UE, the size of one radio record, scores every UE in a report with ONNX Runtime 1.30.0 through its C API, and answers each report with one E2SM-RC control message (radio bearer control, QoS flow mapping, the only control the emulated node implements). Then it waits for the acknowledgment. Because the emulated node reports synthetic values for 0 to 12 UEs, the xApp scores held-out test windows in their place. Before any timing, the C code reproduced the Python scores on all 589 test windows to within 4×10^{-7} .

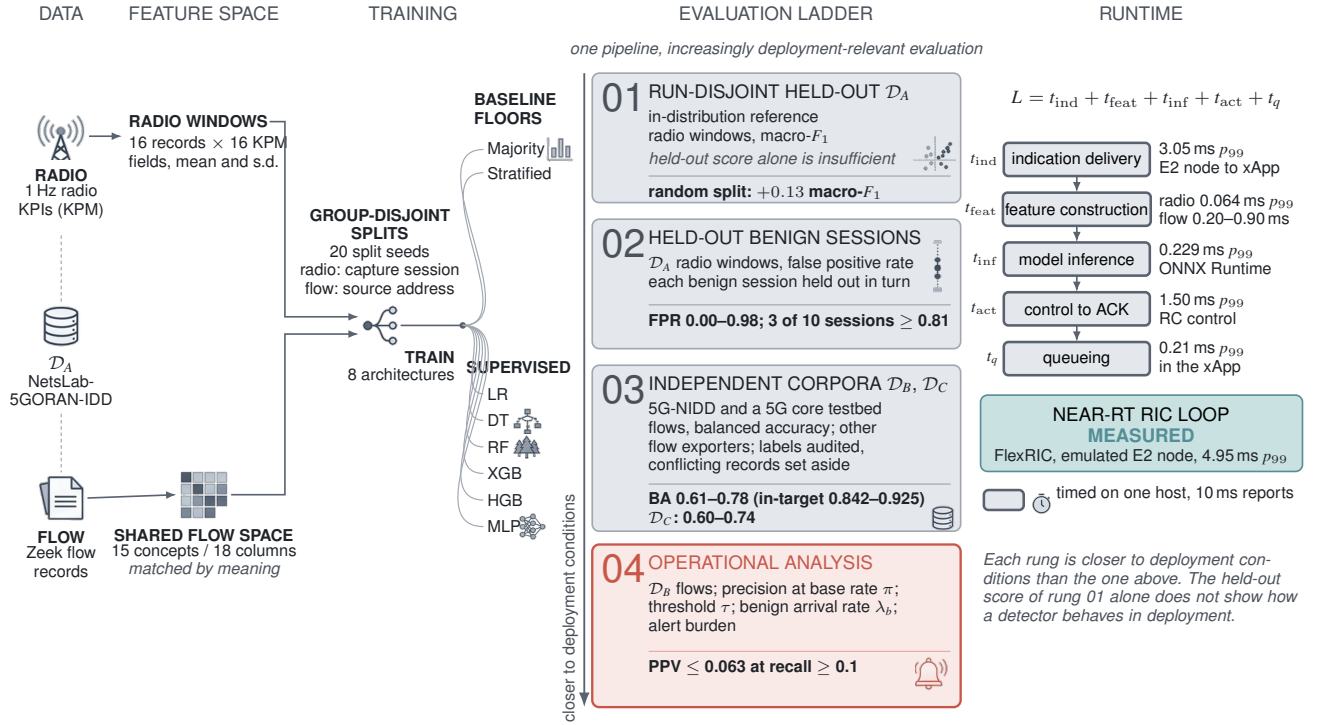


FIGURE 2. Experimental pipeline and evaluation ladder. Radio windows and flows from D_A are summarized separately, split by group, and used to train eight architectures. Each rung of the ladder moves closer to deployment conditions: a run-disjoint held-out split, held-out benign sessions, two independently collected corpora evaluated on flows with consistent labels, and an operational analysis at a declared base rate. The bold line under each rung gives the measured result. The runtime column gives the 99th percentile of each term of Eq. (5), measured on one host through a FlexRIC near-real-time RIC with an emulated E2 node at a 10 ms reporting period. Flow feature construction was timed without a RIC.

Delivery time comes from the timestamp the node writes into the KPM header, which works because all processes share one clock. No radio, no real E2 node, and no container platform took part. All other latency figures were measured without a RIC.

The protocol comparison on the radio layer uses 20 split seeds with two model seeds each, and the other radio-layer and flow-layer experiments use 20 split seeds with one model seed. Calibration uses 10 split seeds of a three-way source split into training, calibration, and test groups. We declare an attack prevalence of $\pi = 0.002$ and a benign arrival rate of $\lambda_b = 240,000$ flows per hour and vary both over a range. On the radio layer, false alerts are counted per benign UE-hour. Neither parameter comes from an operator. Our rate of about 67 flows per second is below the several hundred flows per second that Jin et al. expect at a heavily loaded base station of a large LTE network [61], so the alert volumes we report probably understate those of an edge site serving several cells. Latency is measured per call with a monotonic clock and garbage collection enabled, after 1,000 warm-up calls, over 20,000 calls, with one thread per library. Every quantile comes with a distribution-free confidence interval.

VI. RESULTS

The results follow the evaluation ladder. We start with protocol sensitivity within one corpus and the role of time and benign sessions, then turn to the target corpus and a published

result on it, transfer between corpora and what the gap is made of, and a third corpus. Alert burden, calibration, and latency come last.

A. EVALUATION-PROTOCOL SENSITIVITY

Table 5 and Fig. 3 compare three protocols on the radio layer. Against a run-disjoint split, a random split raises macro- F_1 by 0.09 to 0.17 across the six architectures, 0.13 on average, and the gain is positive in at least 16 of the 20 splits for every architecture. The trivial baselines gain nothing. For the average gain, the corrected interval is $[-0.01, 0.27]$ ($p = 0.06$). With only thirty capture sessions we cannot pin down the size of the effect, and no single architecture's gain survives Holm correction. An uncorrected paired t -test on the same data would have given $p = 6 \times 10^{-5}$.

The gain is not caused by test categories missing from training: in all 20 run-disjoint splits, every test category also appeared in training. The category-stratified protocol guarantees this by design and gives a gain of 0.14 to 0.19. We do not count its larger gain as extra evidence, though, because it pushes sessions of rare categories into the test set and makes that set harder. A model-free probe exposes the mechanism. Under a random split, 68% of the test windows have their nearest training window in the same capture session (65% to 71% across splits), and a one-nearest-neighbor (1-NN) lookup reaches a macro- F_1 of 0.96, against 0.81 under a run-disjoint split (Table 5). Since every radio label is also a session

TABLE 5. Protocol sensitivity on the $\mathcal{D}_A$ radio layer: macro- F_1 under random, run-disjoint (RD) and category-stratified run-disjoint (CS) splits, and paired gains of the random split with Nadeau-Bengio corrected 95% intervals over 20 split seeds. No supervised architecture's gain is significant after Holm correction. 1-NN: a model-free nearest-neighbor lookup on the same splits.

Model	Random	Run-disj.	Cat.-strat.	Δ_{R-RD} [95% CI]	Δ_{R-CS} [95% CI]
LR	0.836	0.744	0.697	+0.093 [-0.05, +0.24]	+0.139 [-0.06, +0.33]
DT	0.950	0.782	0.758	+0.168 [+0.02, +0.32]	+0.192 [+0.02, +0.36]
RF	0.970	0.853	0.816	+0.117 [-0.04, +0.27]	+0.154 [-0.02, +0.32]
XGB	0.977	0.837	0.808	+0.140 [-0.00, +0.28]	+0.169 [-0.02, +0.35]
HGB	0.982	0.844	0.831	+0.138 [-0.01, +0.28]	+0.152 [-0.03, +0.33]
MLP	0.954	0.828	0.809	+0.126 [-0.01, +0.26]	+0.146 [-0.02, +0.31]
Strat.	0.498	0.507	0.500	-0.009 [-0.01, -0.00]	-0.002 [-0.01, +0.01]
Maj.	0.431	0.434	0.447	-0.002 [-0.02, +0.01]	-0.016 [-0.04, +0.01]
1-NN	0.958	0.814	0.794	+0.144 [+0.00, +0.28]	+0.164 [+0.01, +0.31]

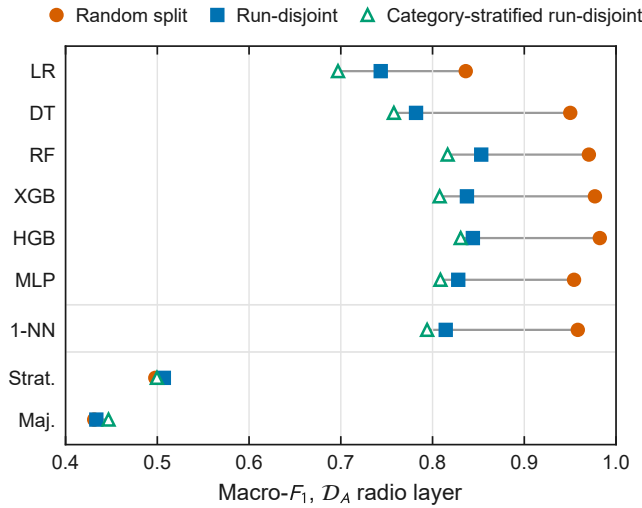


FIGURE 3. Macro- F_1 on the $\mathcal{D}_A$ radio layer under three protocols, mean over 20 split seeds; the gray line spans the three protocols of each model. Every supervised model and the model-free 1-NN lookup score higher under the random split, and the input-blind baselines do not move. Keeping every attack category in training (category-stratified split) does not remove the drop. Corrected intervals are given in Table 5.

label, knowing the session is enough to know the label. A random split lets any model use this, a lookup table included.

Under the run-disjoint split, the lookup beats LR and DT and comes within 0.04 of the best supervised architecture. On this layer, then, the learned models add little to simply retrieving similar windows. The two sequence models in Table 6, which read the raw 16-step KPM sequences instead of window summaries, show the same pattern more sharply. They reach 0.923 to 0.944 under a random split and 0.755 to 0.759 under a run-disjoint split. That gain of 0.16 to 0.19 has corrected intervals that exclude zero for both models (Holm-adjusted $p \leq 0.02$ within the pair).

B. TIME ORDER AND BENIGN SESSIONS

The radio layer spans 52.96 days, but its capture order follows the scenario schedule (Table 7). Nine benign sessions were recorded on days 0 to 6, the five attack categories in blocks on days 47 to 53, and a last benign session on day 53. Each

TABLE 6. Sequence models on the same radio windows and splits as Table 5.

Model	Random	Run-disj.	Cat.-strat.	Δ_{R-RD} [95% CI]
1D-CNN	0.944	0.755	0.763	+0.189 [+0.05, +0.32]
GRU	0.923	0.759	0.769	+0.165 [+0.04, +0.29]

TABLE 7. False positive rate on each benign radio session when that session alone is held out and the models train on all other 29 sessions, mean and range over the six architectures. Day: capture start relative to the first session. Alerts per benign UE-hour: $FPR \times 225$ windows.

Session	Day	UEs	Windows	FPR	Range	Alerts/UE-h
0	0.0	1	57	0.00	[0.00, 0.00]	0
1	0.0	1	54	0.02	[0.00, 0.06]	6
2	2.8	1	58	0.00	[0.00, 0.00]	0
3	2.9	1	55	0.07	[0.02, 0.20]	15
4	3.0	1	24	0.81	[0.54, 0.96]	181
5	3.9	1	56	0.14	[0.00, 0.77]	31
6	4.0	2	60	0.00	[0.00, 0.00]	0
7	4.0	1	95	0.01	[0.00, 0.03]	2
8	6.0	1	56	0.98	[0.93, 1.00]	220
29	52.9	3	146	0.87	[0.84, 0.93]	196

session is benign or belongs to one attack category from start to end. Split the sessions by order, and whole attack categories end up held out. For training fractions between 0.4 and 0.7, the latest sessions contain 1 to 4 attack categories never seen in training, plus a single benign session. What such a split measures is the detection of unseen categories. It is not temporal drift, and we do not report it as a drift estimate.

To hold category coverage fixed, we held out the latest session of every category, the benign category included, and compared the result with 50 control draws that each hold out a randomly chosen session per category (Table 8). Holding out the latest sessions gives a balanced accuracy of 0.52 to 0.57, against control means of 0.75 to 0.83. Yet 12% to 32% of the control draws score just as low. Holding out the earliest sessions gives 0.95 to 0.99, a level that 4% to 34% of the control draws match or beat. Time order, in short, stays within the ordinary variation between sessions.

Much of that variation depends on which benign session ends up in the test set. Held out one at a time (Table 7 and Fig. 4), 6 of the ten benign sessions give a false positive rate of at most 0.07, while sessions 4, 8 and 29 (days 3, 6 and 53) give between 0.81 and 0.98. The last benign session, the only one recorded during the attack period, reaches 0.87 (196 false alerts per benign UE-hour). Two sessions from the first week come close to that. With ten benign sessions, a change over time cannot be told apart from this heterogeneity, and a detector's false positive rate at a new site cannot be predicted from its rate on the benign sessions it was tested on.

C. THE TARGET CORPUS: LABELS AND A PUBLISHED RESULT

Label conflicts. Before running any transfer to $\mathcal{D}_B$, we checked whether its labels follow from its flow records. For half of the corpus they do not. Of the 1,215,889 flows, 52% (632,378) match a flow with the opposite label in every field

TABLE 8. Category-stratified time split on the $\mathcal{D}_A$ radio layer: balanced accuracy when the latest or the earliest session of every category is held out, against the mean of 50 draws that hold out a random session of every category. “% ctrl. $\leq$ ”: share of control draws scoring at or below the time split. Every test category is in training.

Model	Control		Latest held out		Earliest held out	
	BA	% ctrl. $\leq$	BA	% ctrl. $\leq$	BA	% ctrl. $\leq$
LR	0.749	0.573	0.950	96	0.950	96
DT	0.774	0.521	0.962	84	0.962	84
RF	0.811	0.562	0.970	66	0.970	66
XGB	0.812	0.567	0.978	72	0.978	72
HGB	0.817	0.550	0.995	94	0.995	94
MLP	0.827	0.574	0.983	94	0.983	94

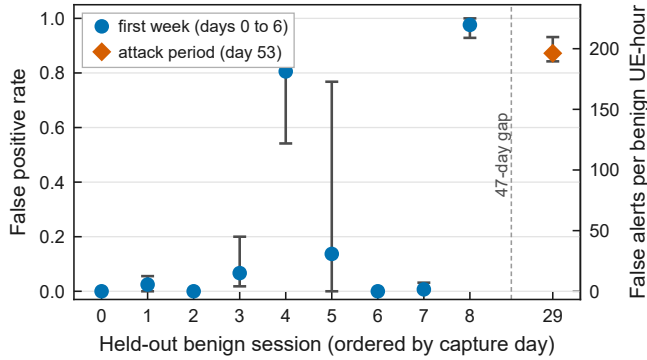


FIGURE 4. False positive rate on each benign radio session when that session alone is held out: mean (marker) and range (bar) over the six architectures, sessions in capture order. The right axis gives the same rate as false alerts per benign UE-hour (225 windows). Within a single week, sessions range from no false positives to almost every window flagged. The benign session matters more than the time of capture.

except the row index, *Seq*, and *Offset* (Table 9, Fig. 5). All of them sit in the two UDP-flood captures, one per base station, where they make up 75% of file 5 and 98% of file 15. The match is exact. Each of the 281,529 flows labeled UDP flood in file 15 has an identical copy labeled benign in file 5, and for 33,704 of the 33,708 distinct records the two counts are equal. The copies amount to 59% of the corpus’s benign class.

Which label is right? The combined files carry no addresses, but the release with all fields preserved does, and it lines up with them row for row. All 281,525 benign copies are UDP flows from 10.155.15.7 to 10.41.150.68. That host pair, protocol, and time window make up the flood labeled as attack in file 15. The attack labeled in file 5 comes from a different host, 10.155.15.4, aimed at the same address. The capture at the first station evidently recorded the second station’s flood too, and only its own attacker got an attack label. We therefore treat the benign copies as mislabeled attack traffic, which makes the evaluation without them the one with correct labels. No label is changed, and every $\mathcal{D}_B$ result is reported with and without the copies.

These conflicts put a ceiling on any classifier that reads the record content. Even evaluated on the very flows it was built from, the best possible function of the 90 content fields reaches a balanced accuracy of only 0.766. In the 18 shared columns the ceiling is 0.753, which is where DT, XGB, HGB,

TABLE 9. Label conflicts in $\mathcal{D}_B$ (5G-NIDD). Content: all fields of the published encoding except row index, *Seq*, *Offset*, and the labels. Ceiling: balanced accuracy of the best lookup on distinct records, in sample.

Features	Cols	Records	% in conflicting records			BA ceiling
			All	Benign	Attack	
Content fields	90	374 946	52.0	58.9	47.5	0.766
Shared space	18	246 281	62.4	59.7	64.1	0.753
Content + <i>Seq</i> , <i>Offset</i>	92	1 215 869	0.0	0.0	0.0	1.000

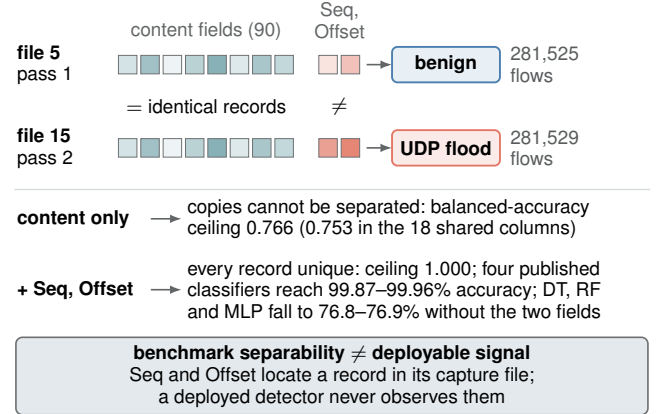


FIGURE 5. Illustration of the label conflict in $\mathcal{D}_B$. Identical flow records occur in two capture files from the two passes and are labeled benign in one file and UDP flood in the other (the tiles are schematic). No function of the record content can separate the copies, which bounds the attainable balanced accuracy. Only the record-position fields *Seq* and *Offset* differ between them. These fields make every record unique and account for the published 99.9%, yet a deployed detector never sees them.

and the MLP land when trained and tested on a random split of $\mathcal{D}_B$ (Table 10). Add *Seq* and *Offset*, and every record becomes unique. The ceiling then rises to 1.000.

References within the target. To tell the cost of transfer apart from the limits of $\mathcal{D}_B$ itself, we also trained and tested the six architectures on $\mathcal{D}_B$ alone (Table 10). A random split gives 0.708 to 0.752 in the shared space, which is the ceiling above, and 0.842 to 0.925 once the benign copies leave the test set. Held-out capture files give 0.931 to 0.982 in the shared space and 0.990 to 1.000 with native features, although none of the three draws held out a UDP-flood capture. Moving between base stations, inside one corpus and with one exporter, drops balanced accuracy to 0.618 to 0.641 from station 1 to station 2 and to 0.639 to 0.650 the other way. The copies cause both drops. Trained on station 1, where the flood copies carry benign labels, a model misses the floods of station 2. Trained on station 2, it flags the benign-labeled copies of station 1, and it scores 0.973 to 0.988 when those copies are removed from the test set. Where labels follow from record content, the shared space separates the classes of $\mathcal{D}_B$ almost perfectly. The target corpus is not what limits transfer.

A published result under the same ladder. The dataset authors report binary accuracies of 99.85% to 99.95% on a random 70/30 split [11]. We rebuilt their pipeline: the published one-hot encoding, the ten features with the highest

TABLE 10. References inside $\mathcal{D}_B$: balanced accuracy of models trained and tested on $\mathcal{D}_B$ only, mean over n draws of a 240,000-flow training sample.
 *Range over the six architectures on the test flows without the benign copies of Table 9. Capture-file draws hold out four of the 20 files.

Features	Split	LR	DT	RF	XGB	HGB	MLP	BA*	n
Shared (18)	Random	0.708	0.752	0.723	0.752	0.752	0.750	0.84–0.92	3
Shared (18)	Capture file	0.955	0.982	0.940	0.943	0.956	0.931	0.93–0.98	3
Shared (18)	Station 1 to 2	0.618	0.638	0.640	0.641	0.640	0.635	0.62–0.64	1
Shared (18)	Station 2 to 1	0.639	0.648	0.648	0.648	0.648	0.650	0.97–0.99	1
Native	Random	0.710	0.764	0.735	0.764	0.764	0.764	0.80–0.87	2
Native	Capture file	0.990	1.000	0.999	1.000	1.000	0.994	0.99–1.00	2
Native	Station 1 to 2	0.698	0.661	0.662	0.662	0.662	0.662	0.66–0.70	1
Native	Station 2 to 1	0.648	0.648	0.649	0.650	0.650	0.654	0.99–1.00	1

ANOVA F -scores, z -score normalization, and their five classifiers, which we ran with scikit-learn defaults because the tuned values are not reported. Our F -scores rank the features in exactly their order, and on their split we obtain 99.87% to 99.96% for DT, RF, KNN, and the MLP (Naive Bayes 92.71%), in line with their results (Table 11). Their two top features are `Seq` and `Offset`, which encode where a record sits in its capture file. No deployed detector can observe them. By the bounds above, they are also exactly what separates the conflicting records.

Swapping only these two fields for the next two in their ranking brings the accuracy of DT, RF, and the MLP down to 76.8% to 76.9% (balanced accuracy 0.704 to 0.706, false positive rate 0.588 to 0.591). The attack information is still there, since RF still reaches 0.9999 on the test flows without the benign copies. The drop comes from the label conflict. Put simply, the two counters let the published models tell identical records apart by their position in the file. With capture files held out, the published features give average balanced accuracies of 0.859 to 0.923 for DT, RF, and the MLP, with RF ranging from 0.82 to 1.00 across folds. Without the counters the average is 0.799 to 0.800, RF ranges from 0.50 to 1.00, and accuracy falls to 1.6% on the fold that trains on one flood capture and tests on the other. Across base stations the published features give 0.762 to 0.895, and the features without counters 0.656 to 0.692. A 99.9% accuracy on this corpus says little about a new capture file or base station. About a deployment, where record positions do not exist, it says nothing.

D. TRANSFER ACROSS CORPORA

Table 12 and Fig. 6 give the results of transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$. Measured in macro- F_1 , every supervised model loses 0.10 to 0.13. So does the majority baseline, which uses no features at all: it loses 0.11, simply because $\mathcal{D}_B$ has a lower attack prevalence than $\mathcal{D}_A$. Net of that baseline, the macro- F_1 loss is -0.018 to +0.015, and none of these losses stays significant after correction. Macro- F_1 is the wrong yardstick here. Balanced accuracy, which prevalence leaves untouched, shows the loss plainly. It falls from 0.81–0.91 on held-out source data to 0.53–0.63 on $\mathcal{D}_B$. This drop of 0.21 to 0.36 is significant for 5 of the six architectures after correction, positive in at least 18 of the 20 splits for every architecture, and 0.28 [0.15, 0.41] on average. The trivial baselines do not

move, and target ROC-AUC lies between 0.53 and 0.70.

The detectors stay above chance on the target, but not by much. Their average target balanced accuracy is 0.57, and the corrected interval excludes 0.5 for 4 of the six architectures (3 after Holm correction). As floors, the stratified sampler scores a macro- F_1 of 0.424 on $\mathcal{D}_B$ and a fair coin would score 0.494. We give both, because a macro- F_1 floor depends on the prior it samples from. On every prevalence-independent measure LR transfers best (balanced accuracy 0.63, ROC-AUC 0.70). The MLP, strongest on held-out source data (0.91), reaches 0.55 on the target. Source performance, in other words, does not predict target performance in a prevalence-independent metric: the rank correlation in balanced accuracy across the six architectures is -0.26 ($p = 0.62$). In macro- F_1 it is 0.83 ($p = 0.04$), but macro- F_1 rankings move with prevalence, and six architectures give little precision to either estimate.

Effect of the target labels. Some of this loss comes from the label conflict of Section VI-C. A detector that learned to recognize a UDP flood on $\mathcal{D}_A$ finds the floods in capture file 15 and, by the same rule, flags 0.00 to 0.85 of the benign-labeled copies in file 5. LR finds neither. With the copies excluded, in a deterministic re-run of 10 of the 20 split seeds, target balanced accuracy is 0.61 to 0.78 (0.74 to 0.78 for the five nonlinear architectures, 0.61 for LR). The loss against held-out source data shrinks to 0.00 to 0.25 and is significant for 1 of the six architectures after correction. The conflicts turn the ranking upside down. LR looks like the best transfer model on all flows and is the worst on the flows with consistent labels. Even on these flows, transfer falls 0.06 to 0.25 short of the in-target reference in balanced accuracy (Table 10).

The reverse direction (Table 13) looks different. Balanced accuracy on $\mathcal{D}_A$ reaches 0.61 to 0.76, and the change against the source reference ranges from -0.037 to +0.145. That reference is optimistic, since it comes from a random split. LR, XGB, HGB, and the MLP lose little or no balanced accuracy, while DT and RF lose a significant amount. Macro- F_1 still falls sharply, because $\mathcal{D}_A$ is 94.63% attack traffic and a handful of false negatives swamps its small benign class. A model trained on the narrower $\mathcal{D}_B$ thus separates the classes of $\mathcal{D}_A$ better than a model trained on $\mathcal{D}_A$ separates those of $\mathcal{D}_B$. We have not tested why. One possibility is that the larger and more varied benign traffic of $\mathcal{D}_B$ teaches a better model of benign behavior.

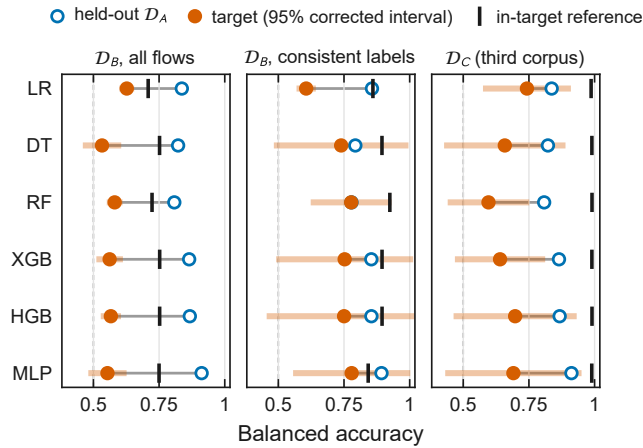
No architecture handles both classes of $\mathcal{D}_B$ acceptably (Ta-

TABLE 11. The 5G-NIDD authors' pipeline [11] reproduced and moved one step per rung: balanced accuracy on $\mathcal{D}_B$. R0 is their protocol. * RF on the test flows without the benign copies of Table 9. KNN is evaluated at R0 only.

Rung	Split	Features	DT	Balanced accuracy				RF	
				RF	KNN	NB	MLP	BA*	FPR
R0	Random 70/30	Published 10	0.9995	0.9996	0.9988	0.9310	0.9986	0.9997	0.0005
R1	Random 70/30	No Seq/Offset	0.7059	0.7059	–	0.6814	0.7042	0.9999	0.5881
R2	Capture file	Published 10	0.8594	0.9226	–	0.7501	0.8669	0.9578	0.0723
R2	Capture file	No Seq/Offset	0.7988	0.7996	–	0.7593	0.8004	0.8958	0.1937
R3	Base station	Published 10	0.7619	0.8949	–	0.6598	0.8577	0.9354	0.0819
R3	Base station	No Seq/Offset	0.6557	0.6555	–	0.6271	0.6919	0.8284	0.3465

TABLE 12. Transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$ in the 18-column shared space, mean over 20 split seeds. Δ_{BA} : source minus target balanced accuracy, with Nadeau-Bengio corrected 95% interval; *: significant after Holm correction. Last column: macro- F_1 loss minus the majority baseline's loss. Italic rows are the trivial baselines.

Model	Macro- F_1		Balanced accuracy			Target	Δ_{F_1} vs. Maj.
	Src.	Tgt.	Src.	Tgt.	Δ_{BA} [95% CI]	ROC-AUC	
LR	0.626	0.530	0.837	0.627	+0.210* [+0.10, +0.32]	0.701	-0.018
DT	0.619	0.518	0.822	0.533	+0.290* [+0.08, +0.50]	0.526	-0.013
RF	0.701	0.572	0.808	0.581	+0.227 [-0.01, +0.47]	0.578	+0.015
XGB	0.646	0.532	0.864	0.562	+0.303* [+0.14, +0.47]	0.572	-0.001
HGB	0.647	0.537	0.867	0.566	+0.301* [+0.13, +0.47]	0.581	-0.004
MLP	0.655	0.531	0.912	0.553	+0.359* [+0.22, +0.50]	0.557	+0.010
<i>Strat.</i>	0.494	0.424	0.502	0.500	+0.001 [-0.01, +0.01]	0.500	-0.044
<i>Maj.</i>	0.492	0.378	0.500	0.500	+0.000 [+0.00, +0.00]	0.500	+0.000

**FIGURE 6.** Balanced accuracy of detectors trained on $\mathcal{D}_A$: on held-out $\mathcal{D}_A$ (open) and on the target (filled, with the Nadeau-Bengio corrected 95% interval), for $\mathcal{D}_B$ on all flows (20 split seeds), $\mathcal{D}_B$ without the benign copies of Fig. 5 (10 seeds), and the third corpus $\mathcal{D}_C$ (20 seeds). The vertical bar is a reference trained and tested inside the target on a random split. Both input-blind baselines sit on the dashed chance line. On every target, the transferred detectors fall short of the in-target reference.

ble 14). LR keeps a benign specificity of 0.98 but catches only 0.25 of the DoS traffic. The tree models and the MLP catch more attacks and flag 0.43 to 0.56 of the benign traffic along the way. In the reverse direction, categories that $\mathcal{D}_B$ lacks are detected unevenly: the ensembles and the MLP recall at most 0.01 of the web attacks, yet brute-force attacks, equally absent from $\mathcal{D}_B$, are recalled at 0.34 to 0.57.

TABLE 13. Reverse direction, $\mathcal{D}_B$ to $\mathcal{D}_A$. The source reference is a random split of $\mathcal{D}_B$, which has no group key, and is therefore optimistic.

Model	Macro- F_1		Balanced accuracy			Target ROC-AUC	Δ_{F_1} vs. Maj.
	Src.	Tgt.	Src.	Tgt.	Δ_{BA} [95% CI]		
LR	0.708	0.563	0.708	0.745	-0.037* [-0.05, -0.02]	0.827	+0.254
DT	0.752	0.343	0.751	0.607	+0.145* [+0.05, +0.24]	0.622	+0.518
RF	0.730	0.337	0.723	0.630	+0.093* [+0.06, +0.13]	0.845	+0.502
XGB	0.753	0.497	0.752	0.764	-0.012 [-0.03, +0.01]	0.845	+0.364
HGB	0.753	0.488	0.752	0.751	+0.001 [-0.05, +0.05]	0.832	+0.373
MLP	0.752	0.470	0.751	0.728	+0.023 [-0.04, +0.09]	0.814	+0.390
<i>Strat.</i>	0.502	0.417	0.502	0.500	+0.003 [+0.00, +0.00]	0.500	+0.194
<i>Maj.</i>	0.377	0.486	0.500	0.500	+0.000 [+0.00, +0.00]	0.500	+0.000

E. COMPOSITION OF THE TRANSFER GAP

Separability. A classifier trained to tell $\mathcal{D}_A$ flows from $\mathcal{D}_B$ flows reaches a balanced accuracy of 0.999 on the 18 shared columns and 0.997 on the nine exporter-robust columns alone. The largest per-feature shifts are in the protocol mix (most $\mathcal{D}_A$ flows are TCP, most $\mathcal{D}_B$ flows UDP), the direction ratios, and the packet counts (Fig. 7). The robust columns shift as much as the segmentation-dependent ones ($\bar{W}_1$ of 0.28 against 0.27). So the corpora differ both in flow segmentation, which a common exporter would remove, and in traffic composition, which no exporter would.

A common exporter. Because the raw captures of $\mathcal{D}_B$ are public, we could test the exporter directly. We re-extracted them with Zeek 8.0.10, the exporter of $\mathcal{D}_A$, and mapped the Zeek records into the shared space with the same function used for $\mathcal{D}_A$. In every capture of $\mathcal{D}_B$, the published attack flows belong to one host pair. Labeling a flow as attack when its host pair is an attack pair of the same capture reproduces 100% of the published labels, so we carried those labels over to the 346,882 Zeek flows (76% attacks). The flood copies

TABLE 14. Per-category detection, mean over 20 split seeds at $\tau = 0.5$. “Ben.” is specificity; other columns are recall. BF: brute force. Categories absent from $\mathcal{D}_B$ can be tested only in the reverse direction.

Model	$\mathcal{D}_A \rightarrow \mathcal{D}_B$			$\mathcal{D}_B \rightarrow \mathcal{D}_A$ (absent from $\mathcal{D}_B$: DDoS, BF, web)					
	Ben.	DoS	Probe	Ben.	DoS	Probe	DDoS	BF	Web
LR	0.98	0.25	0.64	0.71	0.85	0.92	0.96	0.57	0.30
DT	0.44	0.65	0.29	0.82	0.44	0.47	0.48	0.43	0.13
RF	0.46	0.74	0.19	0.89	0.41	0.55	0.49	0.34	0.00
XGB	0.57	0.58	0.17	0.89	0.73	0.88	0.85	0.35	0.01
HGB	0.54	0.63	0.18	0.87	0.72	0.87	0.83	0.34	0.01
MLP	0.46	0.68	0.25	0.86	0.71	0.73	0.81	0.36	0.00

of Section VI-C reappear in the Zeek output (66,033 flows). They are in the captures, not an artifact of the Argus export. We then reran the transfer on the same 10 split seeds as the Argus results without the copies. The trained models are identical, and only the export of the target changes.

The gap does not close. On all flows, balanced accuracy falls from 0.54–0.62 with Argus to 0.43–0.56 with Zeek, and the paired difference over seeds (–0.15 to –0.06) has a 95% interval below zero for 6 of the six architectures. Without the copies, five architectures move by –0.06 to 0.04, none significantly, and LR loses 0.20. Under Zeek the false positive rate rises for every architecture, to 0.13–0.46 without the copies against 0.03–0.18 with Argus. What is left is the difference in traffic composition that the domain classifier and the robust columns had already pointed to.

Model families and adaptation. Table 15 runs the same splits under three modifications, and none of them produces a detector that transfers. Keeping only the nine exporter-robust columns lifts held-out source balanced accuracy to 0.80 to 0.94, but target balanced accuracy drops to 0.48 to 0.53, no better than chance. The columns that do not depend on flow segmentation carry the compositional shift. CORAL, which reads unlabeled target features, gives a target balanced accuracy of 0.47 to 0.60. It helps only the MLP (0.60 against 0.55 without adaptation), and it pushes the target false positive rate up to 0.55 to 0.99. The novelty detectors, trained on benign traffic alone, already fail on held-out source data, because in this feature space floods and scans look like short benign flows.

Cost of the shared space. Projecting $\mathcal{D}_A$ onto the 18 shared columns costs 0.005 to 0.082 of in-distribution balanced accuracy compared with all 46 non-port Zeek features (Table 16). The low source macro- F_1 values in Table 12 reflect the 94.63% attack prevalence of $\mathcal{D}_A$, not weak detectors, since the matching source balanced accuracy is 0.81 to 0.91.

F. A THIRD CORPUS

Do these results hold beyond one pair of corpora? To find out, we added $\mathcal{D}_C$, the flow records of a 5G core testbed [22], as a second target. It changes the site, the flow exporter, and the vantage point together, and unlike $\mathcal{D}_B$ its labels are consistent. Only 3.2% of its flows sit on records that also occur with the opposite label, and the ceiling on balanced accuracy in the shared columns is 0.992. Trained and tested on $\mathcal{D}_C$ itself, the

TABLE 15. Transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$ under alternative feature sets, adaptation, and model families, mean over 20 split seeds. CORAL reads unlabeled target features.

Arm	Model	Src. BA	Tgt. BA	Tgt. AUC	Tgt. FPR
Shared space (18)	LR	0.837	0.627	0.701	0.023
	DT	0.822	0.533	0.526	0.558
	RF	0.808	0.581	0.578	0.541
	XGB	0.864	0.562	0.572	0.429
	HGB	0.867	0.566	0.581	0.463
	MLP	0.912	0.553	0.557	0.539
Exporter-robust subset (9)	LR	0.804	0.510	0.583	0.854
	DT	0.883	0.480	0.488	0.177
	RF	0.894	0.507	0.469	0.206
	XGB	0.925	0.511	0.533	0.089
	HGB	0.939	0.521	0.541	0.090
	MLP	0.928	0.534	0.499	0.039
CORAL (reads target features)	LR	0.836	0.503	0.646	0.902
	DT	0.812	0.500	0.504	0.870
	RF	0.795	0.481	0.515	0.992
	XGB	0.906	0.473	0.478	0.906
	HGB	0.875	0.481	0.484	0.915
	MLP	0.930	0.602	0.632	0.552
Benign-only novelty	IF	0.503	0.492	0.464	0.024
	AE	0.455	0.479	0.484	0.074

TABLE 16. In-distribution cost of the shared space on $\mathcal{D}_A$ flows: all 46 non-port Zeek features against the 18 shared columns, same splits. Loss interval: Nadeau–Bengio corrected, 95%.

Model	BA full	BA shared	Loss [95% CI]	F_1 full	F_1 shared
LR	0.918	0.837	+0.082 [–0.04, +0.20]	0.725	0.626
DT	0.861	0.822	+0.039 [–0.12, +0.20]	0.663	0.619
RF	0.846	0.808	+0.038 [–0.03, +0.11]	0.726	0.701
XGB	0.885	0.864	+0.020 [–0.05, +0.09]	0.677	0.646
HGB	0.893	0.867	+0.026 [–0.11, +0.16]	0.680	0.647
MLP	0.917	0.912	+0.005 [–0.08, +0.10]	0.708	0.655

six architectures reach 0.987 to 0.990 on a random split, and 0.660 to 0.826 when the capture file of one attack type is held out so that this type is never seen in training.

For transfer from $\mathcal{D}_A$ (Table 17, 20 split seeds), balanced accuracy on $\mathcal{D}_C$ ranges from 0.60 to 0.74 and ROC-AUC from 0.70 to 0.93. Every architecture detects the SYN flood, an attack type seen in training. Recall on the ICMP flood ranges from 0.10 to 1.00, and on the control-plane PFCP attack from 0.00 to 0.99. Most errors come from the benign traffic of the new site. The false positive rate is 0.225 to 0.807, and up to 0.81 of the core network’s own signaling is flagged. The loss against held-out source data (0.09 to 0.22) varies so much between split seeds that none of it is significant after

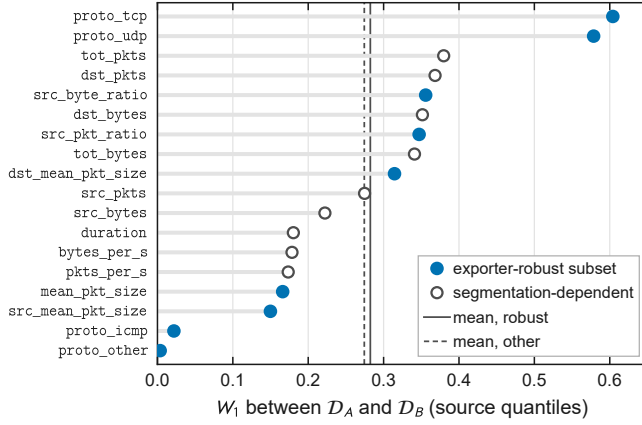


FIGURE 7. Covariate shift between $\mathcal{D}_A$ and $\mathcal{D}_B$ per shared column: first-order Wasserstein distance after the quantile transform fitted on source training data, Eq. (2). Filled markers are the exporter-robust subset, fixed before any result under it was computed; vertical lines are the two group means. The robust columns shift as much as the rest, so the gap is not only a matter of how the two exporters segment flows.

correction. From $\mathcal{D}_B$ (10 split seeds), balanced accuracy spans 0.42 to 0.85, and ROC-AUC drops as low as 0.38. Which architecture transfers best depends on the pair of corpora. None transfers consistently. At $\pi = 0.002$, the best operational precision at any threshold that keeps recall above 10% is 0.092 from $\mathcal{D}_A$ and 0.031 from $\mathcal{D}_B$, and no architecture passes the deployability test in either direction (0 and 0 of six). The picture of Section VI-D returns at a third site, with a third exporter and consistent labels: the SYN flood is caught, and much of the site's benign traffic is not recognized as benign.

G. ALERT BURDEN

Table 18 gives pooled operating points at $\tau = 0.5$ for three evaluation sets, with 95% cluster-bootstrap intervals. On the radio layer, the six detectors have a pooled window-level false positive rate of 0.22 to 0.33 (50 to 74 false alerts per benign UE-hour) and, at $\pi = 0.002$, an operational precision of 0.0055 to 0.0080, against a corpus precision of 0.90 to 0.93. These point estimates are unstable. The false positive rate depends on which of the ten benign sessions enter the test set (Table 7). The bootstrap interval of the pooled rate spans 0.01 to 0.65 across architectures (3 to 146 false alerts per benign UE-hour), and the interval for operational precision at $\tau = 0.5$ spans 0.0028 to 0.13. Thirty sessions cannot fix the alert burden of the radio layer. All they show is that it depends on the benign conditions a detector happens to meet.

The flow layer tells a similar story. On held-out $\mathcal{D}_A$ flows the false positive rate is 0.066 to 0.201, which at $\lambda_b = 240,000$ benign flows per hour means 15,726 to 48,289 false alerts per hour. On $\mathcal{D}_B$ it is 0.023 to 0.558, or 5,620 to 133,990 false alerts per hour. Without the benign copies of Section VI-C it is 0.026 to 0.183. That still comes to 6,158 to 44,003 false alerts per hour, at an operational precision of 0.0072 to 0.0398.

Raising the decision threshold improves precision only on held-out $\mathcal{D}_A$ flows (Fig. 8). Because the best threshold is

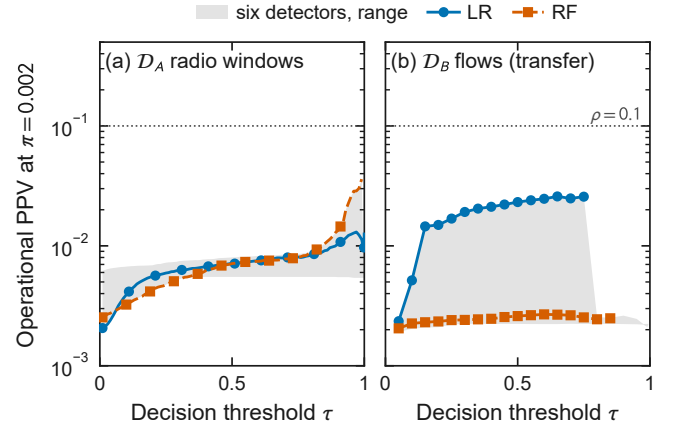


FIGURE 8. Pooled operational precision at $\pi = 0.002$ against the decision threshold on (a) the $\mathcal{D}_A$ radio layer and (b) $\mathcal{D}_B$ flows after transfer, at thresholds that keep recall at or above 0.10. The gray band spans the six architectures. The dotted line is the precision bound $\rho = 0.1$ of the deployability test, and no threshold reaches it on either evaluation set.

picked on the evaluation data, the following values are upper bounds. With recall kept above 10%, high thresholds reach an operational precision of 0.42 on held-out $\mathcal{D}_A$ flows. On $\mathcal{D}_B$, the best any detector achieves is 0.026, or 0.063 without the copies. The radio layer's best pooled value is 0.036, but for 4 of the six architectures its bootstrap interval reaches 1.0: a resample that leaves out the benign sessions flagged almost entirely leaves thresholds with no false positive at all. How corpus precision relates to operational precision is pure arithmetic in π . Moving π from 10^{-4} to 0.5 changes the ratio between them from 2,627 to 1.18 (Fig. 9). An operational precision means something only when the assumed prevalence is stated. *Estimation of operational precision.* Near zero false positives, PPV is strongly nonlinear, so a fold that happens to contain no false positive yields $\text{PPV} = 1$ whatever its recall. Averaging PPV over folds rewards the architecture that happens to draw the most such folds. On the radio layer, the same counts put the ratio between the best and the worst architecture at 1.45 when pooled, 1.89 as a median over folds, and 16.0 as a mean over folds (Table 19). We use pooled counts throughout.

H. CALIBRATION AND PRIOR SHIFT

Under transfer the scores are poorly calibrated (Fig. 10). We fitted three calibrators on a group-disjoint calibration fold from the source (Table 20). Whether a calibrator can move the achievable operating points depends on whether it keeps the order of the scores. Temperature scaling does, and it changes ROC-AUC by at most $2e-05$. Platt scaling is monotone too, but its fitted slope can be negative, and for DT it inverted the ranking in 1 of 10 seeds, shifting ROC-AUC by 0.95. In that seed the tree's scores carried no ranking information on the calibration fold (ROC-AUC 0.48 on 96 held-out source addresses) even though they ranked the test fold well (0.97). Platt scaling fitted the calibration fold correctly, with a slope of -0.55, and in doing so inverted the ranking on the test

TABLE 17. Transfer to the third corpus $\mathcal{D}_C$ (5G core testbed, NFStream flows [22]): balanced accuracy on held-out source data and on $\mathcal{D}_C$, target ROC-AUC, recall per attack type, and the share of the core's background signaling flagged (Bkg.). Mean over 20 split seeds from $\mathcal{D}_A$ and 10 from $\mathcal{D}_B$, at $\tau = 0.5$.

Model	Src. BA	Tgt. BA	Trained on $\mathcal{D}_A$					Trained on $\mathcal{D}_B$	
			AUC	SYN	ICMP	PFCP	Bkg.	Tgt. BA	AUC
LR	0.837	0.742	0.800	1.00	0.10	0.00	0.27	0.517	0.726
DT	0.822	0.658	0.696	1.00	1.00	0.79	0.67	0.422	0.384
RF	0.808	0.596	0.926	1.00	1.00	0.99	0.81	0.540	0.697
XGB	0.864	0.640	0.888	1.00	1.00	0.74	0.70	0.688	0.814
HGB	0.867	0.698	0.932	1.00	1.00	0.80	0.61	0.565	0.632
MLP	0.912	0.691	0.837	1.00	0.20	0.61	0.44	0.848	0.766
<i>Maj.</i>	0.500	0.500	0.500	1.00	1.00	1.00	1.00	0.500	0.500

TABLE 18. Pooled operating points at $\tau = 0.5$, 20 split seeds. FPR interval: 95% cluster bootstrap resampling split seeds and clusters (radio sessions, $\mathcal{D}_A$ source addresses, $\mathcal{D}_B$ capture files; for flows from a deterministic re-run on 10 seeds, EXP-056, on which the “w/o conflicts” block is also computed). “w/o conflicts”: $\mathcal{D}_B$ without the benign flows whose record also occurs with an attack label. PPV at $\pi = 0.002$. [†]False alerts per benign UE-hour (radio, 225 windows per UE-hour) or per hour at $\lambda_b = 240,000$ benign flows (flows). [‡]Highest pooled PPV over all thresholds with recall at least 0.10.

Corpus	Model	TPR	FPR [95% CI]	Corpus P	PPV	False alerts [†]	Best PPV [‡]
$\mathcal{D}_A$ radio	LR	0.798	0.223 [0.017, 0.547]	0.921	0.0071	50	0.0130
	DT	0.910	0.329 [0.086, 0.639]	0.900	0.0055	74	0.0055
	RF	0.954	0.267 [0.024, 0.650]	0.921	0.0071	60	0.0360
	XGB	0.931	0.251 [0.013, 0.606]	0.924	0.0074	57	0.0116
	HGB	0.945	0.262 [0.018, 0.615]	0.922	0.0072	59	0.0086
	MLP	0.895	0.222 [0.019, 0.560]	0.929	0.0080	50	0.0109
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.766	0.0020	225	0.0020
$\mathcal{D}_A$ network	LR	0.892	0.201 [0.126, 0.445]	0.992	0.0088	48 289	0.0297
	DT	0.852	0.157 [0.068, 0.498]	0.994	0.0108	37 658	0.0120
	RF	0.920	0.198 [0.121, 0.611]	0.993	0.0092	47 568	0.4172
	XGB	0.884	0.090 [0.027, 0.373]	0.997	0.0193	21 625	0.2735
	HGB	0.881	0.066 [0.031, 0.112]	0.998	0.0262	15 726	0.2727
	MLP	0.905	0.071 [0.043, 0.349]	0.997	0.0250	16 996	0.2189
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.967	0.0020	240 000	0.0020
$\mathcal{D}_B$ (transfer)	LR	0.277	0.023 [0.009, 0.079]	0.948	0.0232	5 620	0.0258
	DT	0.624	0.558 [0.111, 0.783]	0.633	0.0022	133 990	0.0022
	RF	0.703	0.541 [0.121, 0.796]	0.667	0.0026	129 843	0.0027
	XGB	0.552	0.429 [0.013, 0.687]	0.666	0.0026	102 890	0.0026
	HGB	0.596	0.463 [0.031, 0.699]	0.665	0.0026	111 104	0.0026
	MLP	0.646	0.539 [0.030, 0.745]	0.649	0.0024	129 320	0.0026
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.607	0.0020	240 000	0.0020
$\mathcal{D}_B$ w/o conflicts	LR	0.269	0.057 [0.038, 0.086]	0.947	0.0094	13 629	0.0102
	DT	0.662	0.183 [0.103, 0.294]	0.931	0.0072	44 003	0.0077
	RF	0.719	0.165 [0.118, 0.258]	0.943	0.0087	39 561	0.0629
	XGB	0.531	0.026 [0.012, 0.047]	0.987	0.0398	6 158	0.0631
	HGB	0.548	0.047 [0.030, 0.080]	0.978	0.0227	11 315	0.0587
	MLP	0.612	0.053 [0.026, 0.102]	0.977	0.0225	12 794	0.0306
	<i>Maj.</i>	1.000	1.000 [1.000, 1.000]	0.790	0.0020	240 000	0.0020

TABLE 19. Operational precision on the radio layer at $\tau = 0.5$, $\pi = 0.002$, estimated three ways from the same 20 folds.

Model	Folds with FP = 0	Pooled	Median	Mean
LR	1	0.0071	0.0118	0.0815
DT	0	0.0055	0.0070	0.0137
RF	1	0.0071	0.0092	0.0814
XGB	4	0.0074	0.0095	0.2185
HGB	2	0.0072	0.0091	0.1291
MLP	0	0.0080	0.0132	0.0476

fold. Small group-disjoint calibration folds, which need not resemble the test fold, are to blame here, not Platt scaling. Isotonic regression is monotone but piecewise constant. It merges scores into ties and, on its fitting data, yields the

ROC convex hull [52], and its largest change in ROC-AUC ranged from 0.04 to 0.21 across architectures. No calibrator lifts the target macro- F_1 above that of the raw scores for any architecture. Fitted on a source with 94.63% attacks, they pull the scores toward that prior and flag more benign traffic on $\mathcal{D}_B$.

The usual remedy for prior shift is to rescale the scores to the target prior [48], which a deployment does not know. Expectation maximization [48] and black-box shift estimation [49] estimate it from unlabeled data under the assumption that only the class balance changes. That assumption fails here. For a true target prior of 0.607, the mean estimates per architecture run from 0.16 to 0.86, and 41% of the individual estimates fall below 0.05 or above 0.95. Even an oracle,

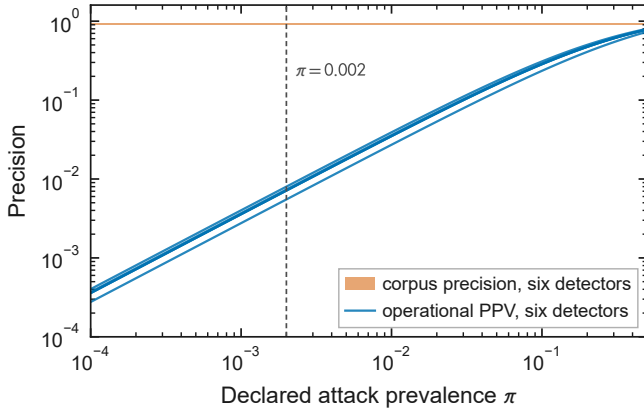


FIGURE 9. Pooled operational precision at $\tau = 0.5$ against the declared attack prevalence π for the six radio-layer detectors (one line each), with their corpus precision as a band. The dashed rule marks $\pi = 0.002$. Operational precision is computed from the same confusion counts with Eq. (4), and only the declared prevalence is varied.

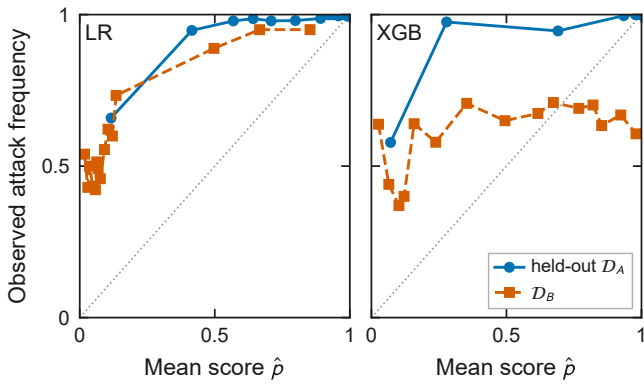


FIGURE 10. Reliability diagrams for LR and XGB on held-out $\mathcal{D}_A$ and on $\mathcal{D}_B$, fifteen equal-mass bins pooled over 20 split seeds.

rescaling with the true prior, does not beat the raw scores (Table 20). The detectors are limited by discrimination on the target (ROC-AUC 0.53 to 0.70), not by calibration.

I. DECISION LATENCY

Table 21 gives per-call latency on one host for three implementations of each model. The first calls scikit-learn with a one-row pandas DataFrame, as our experiment code does. The second calls scikit-learn with a NumPy array on one thread. The third runs ONNX Runtime on one thread and is used only where the exported model reproduces the decisions of scikit-learn (12 of 12 exports, decision agreement at least 1.0000). A call that does nothing has a 99th percentile of 0.001 ms, too little to distort the comparisons.

Implementation matters more than architecture (Fig. 11). For LR, the DataFrame path takes 0.44 ms at the median against 0.11 ms for an array, and input validation accounts for the difference. RF, asked to predict one row through its default thread pool, takes 29 ms at the median, compared with 5.6 ms on one thread and 0.012 ms in ONNX Runtime. On the radio layer the 99th percentile under ONNX Runtime lies

between 0.014 and 0.035 ms for all architectures, while scikit-learn with arrays reaches up to 21.5 ms. Rank architectures by latency and you mostly rank implementations. Numbers from other hosts or languages [21] tell us nothing about the architectures for the same reason. For context only, Obiwewe et al. report inference in the microsecond range for logistic regression and a small MLP inside an OpenAirInterface and FlexRIC near-real-time RIC [21], within an order of magnitude of our ONNX Runtime figures.

A window-level detector aggregates a window and then runs inference. Aggregating one 16-record window takes 0.064 ms at the 99th percentile, and across the architectures with a verified export the upper bound of the 99th percentile of aggregation plus ONNX inference is at most 0.12 ms. On this host, those two terms alone would fit any budget in the specified range. We repeated the radio measurement on the same host and recorded every call for Fig. 11. The medians came out at 1.02 to 2.1 times those in Table 21 (median ratio 1.46). Without CPU isolation, per-call latency depends on the state of the host as well as on the model. The upper 95% bound of the 99th percentile of aggregation plus ONNX inference nonetheless stayed at or below 0.36 ms in the repeat.

Table 22 adds the remaining terms of Eq. (5), measured in the FlexRIC loop of Section V. The six models ran with a report every 10 ms (5,000 reports each after 200 warm-up reports), and LR and RF also with a report every 1 ms (20,016 reports each after 1,000 warm-up reports). Reports carried 7.7 UEs on average, and the xApp scored all of them. At the 10 ms period, a report takes 1.64 ms at the median to get from the E2 node to the xApp, and the control round trip takes 0.78 ms. Inference in the same loop needs at most 0.229 ms at the 99th percentile. For every model, delivery and control account for at least 86% of the mean loop time. Once a RIC is in the path, the choice of model barely matters. The full loop, from report to acknowledged control message, takes 4.45 to 4.95 ms at the 99th percentile, with an upper 95% bound of at most 5.01 ms. That meets the smallest specified budget of 10 ms even at the 99.9th percentile (at most 5.71 ms). The largest single value we saw at this period was 11.2 ms.

Oddly, the loop ran faster at the 1 ms period. Its 99th percentile fell to 1.42 to 1.69 ms, the xApp never queued more than 4 reports, and no run dropped a report. Every term got faster, inference included (at most 0.050 ms at the 99th percentile). We did not control CPU power management, and a host that idles between sparse reports and pays to wake up would produce exactly this pattern. Keep in mind that the loop runs on one host, with an emulated node and loopback SCTP. It bounds the processing cost of the RIC software path. The transport delay of a real deployment is not in it.

For a flow-level detector the expensive part is feature construction, not inference. On the three DDoS captures of $\mathcal{D}_A$, the vectorized exporter produces flow records at 0.20 to 0.90 ms per flow, 22.8 to 41.6 times faster than the per-packet reference implementation on the same captures (4.49 to 37.34 ms per flow), and the cost per flow grows with the number of packets in the flow. On the 2 captures compared

TABLE 20. Calibration over 10 split seeds. Left: largest absolute change in ROC-AUC caused by each calibrator, over seeds and both evaluation sets. Right: target macro- F_1 at $\tau = 0.5$ for raw scores, Platt and isotonic calibration, and Platt followed by prior correction with the EM estimate or the true (oracle) target prior.

Model	max $ \Delta \text{ROC-AUC} $			Target macro- F_1 at $\tau = 0.5$				
	Temp.	Platt	Iso.	Raw	Platt	Iso.	+EM	+Oracle
LR	$< 10^{-6}$	1×10^{-6}	0.209	0.538	0.415	0.391	0.327	0.478
DT	2×10^{-5}	0.946	0.170	0.538	0.395	0.405	0.447	0.436
RF	$< 10^{-6}$	$< 10^{-6}$	0.050	0.579	0.504	0.470	0.440	0.488
XGB	$< 10^{-6}$	$< 10^{-6}$	0.150	0.555	0.450	0.394	0.456	0.518
HGB	$< 10^{-6}$	$< 10^{-6}$	0.095	0.549	0.429	0.399	0.479	0.507
MLP	2×10^{-5}	2×10^{-5}	0.045	0.578	0.405	0.408	0.475	0.534

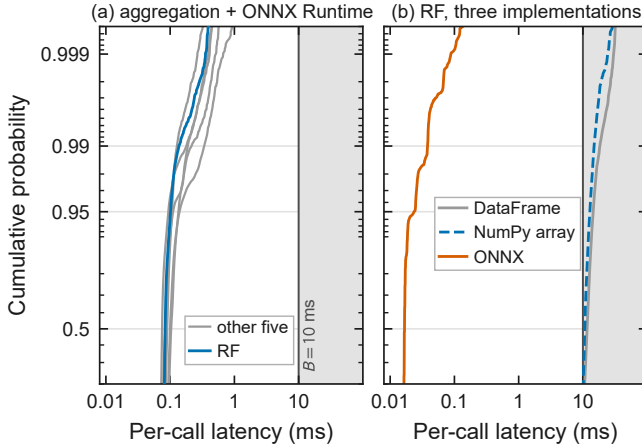


FIGURE 11. Empirical distribution of per-call latency for the radio layer on one host, 20,000 calls per stage, on a logit scale so that the median, 95th, and 99th percentiles are all visible. (a) The decision path of the deployability test, window aggregation followed by ONNX Runtime, for the six architectures. (b) One architecture, RF, under the three implementations of Table 21. The shaded region lies beyond $B = 10$ ms, the smallest budget in the specified range. Only feature construction and inference are timed here. Table 22 gives the full loop through a RIC. The curves come from a repeat of the measurement that kept every call (EXP-060); Table 21 reports the original run.

record by record, the two exporters agree exactly in packets and bytes on all 38,032 shared IPv4 flow keys. The reference implementation also emits 22 and 41 IPv6 link-local keys, which the vectorized exporter skips because it parses IPv4 only. Mapping one exported record onto the shared space with our pandas code costs another 8.5 ms at the median, whereas ONNX inference on flow features needs at most 0.037 ms at the 99th percentile. These numbers describe a Python pipeline on one host. They place the cost in feature construction but cannot tell us what a native exporter at the user plane would cost.

VII. DISCUSSION

A. IMPLICATIONS FOR O-RAN PRACTICE

Data path. E2SM-KPM delivers aggregated measurements once per reporting period. It carries no packets. A window-level detector can live inside the RIC and work on KPM reports. A flow-level detector cannot: it needs a flow exporter near the user plane, in the central unit or at the user-plane function, plus some way to move its records to the xApp,

such as a custom E2 service model or the non-real-time data collection path. That transport adds to t_{ind} in Eq. (5). Our RIC loop measures t_{ind} for KPM reports only and says nothing about the flow layer.

Detection delay. Computation is a small part of the time it takes to detect an attack. A window-level decision has to wait for the end of its window (16 s here). A flow-level decision has to wait until the flow record is exported, which for a long connection means waiting for an idle or active timeout. Anyone reporting computational latency should report the KPM reporting period and the flow timeouts alongside it.

Acting on alerts. An E2SM-RC action such as isolating or steering a UE hits a falsely accused UE just as hard as a guilty one. At the false positive rates in Table 18, automatic mitigation on the radio layer would act against every benign UE 50 to 74 times an hour. Alerts of this quality can inform an operator, or a non-real-time policy through A1. They should not drive automatic control.

Admission and monitoring. The deployability test of Eq. (6) could serve as an admission check before an xApp is switched on at a new site, provided labeled benign and attack traffic from that site is available. After admission, unsupervised shift detection [53] can flag the moment incoming traffic departs from the training distribution. The corpora we studied can be told apart with a balanced accuracy of 0.999, so a monitor of this kind would fire at once.

B. APPLYING THE DEPLOYABILITY TEST

We instantiate Eq. (6) with the bounds below. No O-RAN specification defines them, so we chose them ourselves, and an operator would set them from its own traffic and staffing. The generalization bound is $\delta = 0.10$ in balanced accuracy, applied to the upper end of the corrected 95% interval. The alert term asks for a threshold that detects at least half of the attacks ($r_{\min} = 0.5$), at which at least one alert in ten is real ($\rho = 0.1$ at $\pi = 0.002$), and which raises no more than $A_{\max} = 200$ false alerts per hour at $\lambda_b = 240,000$ benign flows per hour. All alert conditions are checked on $\mathcal{D}_B$, since a detector is deployed on the target and not on the source. For latency we take the strictest budget the specification allows, $B = 10$ ms, and evaluate it on the FlexRIC loop at a 10 ms reporting period. There the upper bound of the 99th percentile is at most 5.01 ms for every architecture (Section VI-I). On

TABLE 21. Per-call latency in milliseconds on one host (Section V): 20,000 calls after 1,000 warm-up calls, one thread per library, garbage collection enabled. No RIC in the path (Table 22 gives the loop through a RIC). Radio path $\bar{p}_{99}$: upper 95% bound of the 99th percentile of window aggregation plus ONNX inference. Dashes: no export that reproduced scikit-learn's decisions.

Model	sklearn, DataFrame		sklearn, array		ONNX Runtime		Radio path $\bar{p}_{99}$	Flow layer, p_{99}	
	p_{50}	p_{99}	p_{50}	p_{99}	p_{50}	p_{99}		array	ONNX
LR	0.44	0.72	0.11	0.17	0.010	0.023	0.12	0.16	0.025
DT	0.31	0.52	0.03	0.04	0.010	0.014	0.11	0.06	0.024
RF	6.25	12.99	5.57	21.50	0.012	0.024	0.10	4.45	0.029
XGB	0.73	1.00	0.16	0.32	0.011	0.021	0.09	0.40	0.024
HGB	1.48	2.72	1.17	1.92	0.013	0.035	0.12	1.22	0.034
MLP	0.43	0.67	0.11	0.21	0.017	0.035	0.11	0.15	0.037

TABLE 22. Decision loop through a FlexRIC near-real-time RIC with an emulated E2 node on one host, in milliseconds, as ranges over models, at a reporting period of 10 ms or 1 ms. Terms follow Eq. (5): delivery from the E2 node to the xApp (t_{ind}), queueing in the xApp (t_q), window aggregation (t_{feat}) and ONNX Runtime inference (t_{inf}) for one UE, and the control message up to its acknowledgment (t_{act}). "Loop" runs from the timestamp in the KPM header to the acknowledgment, for the first UE of a report or for all its UEs.

Term	10 ms, six models		1 ms, LR and RF	
	p_{50}	p_{99}	p_{50}	p_{99}
t_{ind}	1.51–1.86	3.02–3.05	0.51–0.53	0.84–0.98
t_q	0.063–0.071	0.18–0.21	0.015–0.016	0.069
t_{feat}	0.001	0.002–0.003	<0.001	0.001
t_{inf}	0.050–0.12	0.13–0.23	0.014–0.025	0.038–0.050
t_{act}	0.72–0.87	1.40–1.50	0.20–0.21	0.53–0.60
Loop, 1 UE	2.38–2.97	4.36–4.59	0.76–0.79	1.38–1.59
Loop, all	2.48–3.18	4.45–4.95	0.77–0.86	1.42–1.69

one host and with an emulated E2 node, the latency term holds for any budget in the specified range.

Table 23 gives the verdict: no architecture passes (0 of six). All of them fail the generalization term, with upper bounds of 0.32 to 0.50, and all fail the alert term by a wide margin. At a recall of at least 0.5, the best operational precision on $\mathcal{D}_B$ is 0.0027, and the fewest false alerts per hour are 95,507. The verdict is robust to the choice of bounds. It holds for any ρ above 0.0027, any A_{max} below 95,507 per hour, and any δ below 0.32, and the latency term plays no part in it. Neither does the declared prevalence. The false-alert count does not involve π and exceeds A_{max} at every threshold, while the precision condition on its own would need $\pi \geq 0.076$, 38 times the declared value. The label conflicts of Section VI-C do not change the verdict either. On $\mathcal{D}_B$ without the benign copies, 0 of six architectures pass, the best operational precision at a recall of at least 0.5 is 0.0419, and the fewest false alerts per hour are 5,603. The precision condition alone would hold there from $\pi = 0.005$, but the false-alert count still rules out every architecture.

How far would the bounds have to move? Fig. 12 shows it. An architecture meets the alert term only if an operator accepts several thousand false alerts per hour at a precision of a few percent, and it meets the generalization term only if δ exceeds 0.32. Latency would matter only once the first two terms were met, and in our loop it clears the smallest specified budget by about a factor of two.

Attainability of the test. A test that nobody passes may

TABLE 23. Deployability test of Eq. (6) per architecture. $\bar{\Delta}_{BA}$: upper 95% bound of the balanced-accuracy loss (pass if ≤ 0.10). Best PPV and minimum false alerts per hour over thresholds with recall ≥ 0.5 on $\mathcal{D}_B$ (pass if PPV ≥ 0.1 and alerts ≤ 200). B_{min} : smallest budget of the series 1, 2, 5, 10 ms met by the upper 95% bound of the 99th percentile of the FlexRIC loop (all UEs of a report, 10 ms period, emulated E2 node).

Model	$\bar{\Delta}_{BA}$	Gen.	Best PPV	Min. alerts/h	Alert	B_{min} (ms)	$\mathcal{P}$
LR	0.32	×	0.0024	166 328	×	5	0
DT	0.50	×	0.0022	118 342	×	5	0
RF	0.47	×	0.0027	98 410	×	5	0
XGB	0.47	×	0.0026	95 507	×	5	0
HGB	0.47	×	0.0026	99 277	×	10	0
MLP	0.50	×	0.0026	105 744	×	5	0

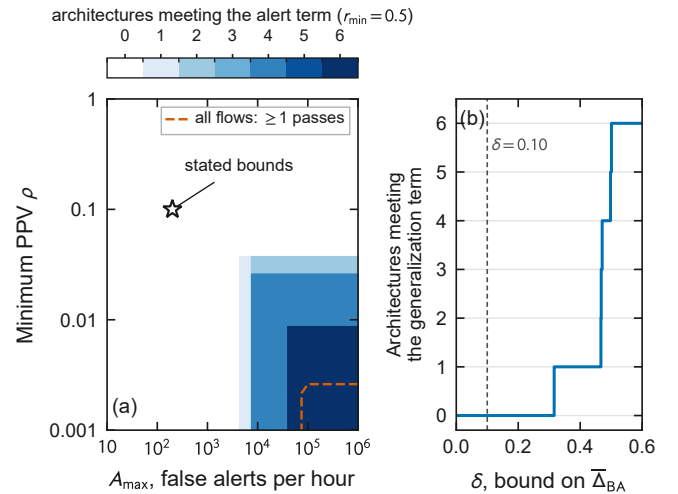


FIGURE 12. Sensitivity of the deployability test to its bounds. (a) Number of the six architectures meeting the alert term ($r_{min} = 0.5$, $\pi = 0.002$, $\lambda_b = 240,000$ flows per hour) on $\mathcal{D}_B$ without the benign copies, as the alert bound A_{max} and the precision bound ρ vary; on all flows, at least one architecture passes only below and to the right of the dashed line. The star marks the bounds we use. (b) Number of architectures meeting the generalization term as δ varies. A detector passes the test only if both terms are satisfied.

just be too strict. The published 5G-NIDD pipeline of Section VI-C suggests otherwise. On its own random split, with the alert term evaluated on that split, 2 of its five classifiers pass, reaching an operational precision of up to 0.79 at $\pi = 0.002$ with as few as 128 false alerts per hour at $\lambda_b = 240,000$. The same pipeline fails one rung later, once the two position counters are removed, and at every rung after that, on all flows and on the flows without the benign copies

alike (Table 11). The test can only be as informative as the evaluation it is applied to. On a random split that still contains record positions it merely confirms that split, and the pass vanishes as soon as the evaluation moves one step closer to deployment. Under the evaluations closest to deployment that the available corpora allow, the detectors in this paper fail.

C. RELATION TO REPORTED RESULTS

Most published results for O-RAN and 5G intrusion detection are in-distribution scores on corpora other than ours, and under a random split our own in-distribution scores are just as high (Table 5). Setting our numbers beside theirs would compare evaluation protocols, not detectors. What our results can say is which of the commonly reported quantities predict deployment. Held-out macro- F_1 under a random split does not. The quantities listed in the reporting requirements of Section X are the ones that actually changed our conclusions.

VIII. REPRODUCIBILITY AND ETHICS

The code, the configuration, the seeded split procedures (every split can be regenerated deterministically), the raw per-seed results, and the generators for every table, figure, and in-text number are available at <https://github.com/adeliusa486/oran-ids-deployability>. One script regenerates every number, table, and figure of this paper from the per-seed results and checks them against the manuscript. The same script can re-run each experiment from the raw corpora after verifying their files by SHA-256, and `REPRODUCE.md` in the repository walks through the procedure. The RIC loop needs a Linux host or WSL2, and the repository builds FlexRIC, the xApp, and Zeek in user space without administrator rights. All three corpora are public. $\mathcal{D}_A$ is distributed by its authors under CC-BY-4.0 [39]. $\mathcal{D}_B$ is distributed by its authors under CC-BY-4.0 [37], including the release with all fields preserved and the GTP-removed captures we used for the address audit and the Zeek re-extraction. $\mathcal{D}_C$ is available from its authors' repository [22], which states no license, so we do not redistribute it. All traffic comes from testbed devices and test networks, not from subscribers, and no personal data was processed. Showing when detectors fail carries a modest dual-use risk. We judge it smaller than the benefit to operators of knowing that reported accuracy need not carry over to their networks.

IX. LIMITATIONS

Flow exporters. Three different tools produced $\mathcal{D}_A$, $\mathcal{D}_B$, and $\mathcal{D}_C$ (Zeek, Argus, and NFStream). We re-extracted $\mathcal{D}_B$ with Zeek, and the gap between $\mathcal{D}_A$ and $\mathcal{D}_B$ did not close (Section VI-E). The Zeek version and configuration behind $\mathcal{D}_A$ are undocumented, though, so our Zeek export of $\mathcal{D}_B$ may still differ from it in timeouts or in its handling of single-packet flows. The Zeek flows take their labels from the published ones through the host pairs of each capture. That rule reproduces every published label on the Argus flows, but a Zeek flow whose host pair carries both attack and benign traffic within one capture would receive the attack label. $\mathcal{D}_C$ was not re-extracted.

Target labels. 59% of the benign flows in $\mathcal{D}_B$ are exact copies of records labeled UDP flood elsewhere in the corpus (Section VI-C). Their addresses mark them as the flood of the attacker at the other base station. The dataset authors have not confirmed this, and until they do, every operating point on $\mathcal{D}_B$ appears with and without these flows. The capture files themselves were recovered from the restarts of `Offset`. The per-station files of the release confirm how the two passes map onto the base stations, since each of the 10 files of the first station matches exactly one capture file of the first pass in row count and attacking host (10 of 10).

Radio capture schedule. Nine of the ten benign radio sessions were recorded six weeks before the first attack session, mostly with a single UE, and every radio label is a session label. A radio detector may thus learn to tell capture periods apart as well as attacks. The one benign session recorded during the attack period is flagged almost entirely. So are two early sessions (Table 7), however, and ten benign sessions are too few to decide between the two explanations.

Number of corpora. Three corpora are enough to show that the transfer gap recurs, not to describe how it varies across deployments. The third comes from a 5G core testbed rather than an O-RAN deployment, contains only flood attacks and one control-plane attack, and consists of three capture files, too few for cluster-based intervals. We found no other O-RAN corpus with paired radio telemetry.

Latency. Every latency figure comes from one workstation without CPU isolation. The RIC loop ran FlexRIC's software path over loopback SCTP to an emulated E2 node with synthetic KPM values. There was no radio, no real E2 node, and no network between sites. What the loop measures is the processing cost of the RIC path, not the delay of a deployment. Because it ran faster at a 1 ms than at a 10 ms reporting period, its figures evidently also depend on the host's power management. CPU and memory utilization under load were not measured.

Operational parameters and the deployability test. The prevalence $\pi = 0.002$ and the benign arrival rate λ_b are declared, not measured on an operational network. We vary both, and operational precision depends on π alone. The bounds of the deployability test are our own choice (Section VII-B). Fig. 12 shows the verdict under other bounds, but an operator may weigh recall, precision, and alert volume differently, and the test ignores detection delay and the cost of a missed attack.

Sample size. The radio layer has 30 capture sessions with one scenario each, so session and attack scenario are confounded, and the protocol and temporal analyses can detect only large effects. With just ten benign sessions, the radio false positive rate, and with it the operational precision, carries a wide interval (Section VI-G). The transfer statistics vary the training set over 20 seeds but always test on one fixed target corpus. They do not capture the variability across other deployments.

Scope. We study binary detection only. Multi-class attribution, feature attribution [62], and the fusion of radio and

flow features remain open. Because the published summary files cannot be joined record by record, fusion needs the raw archives, as in [31]. Three attack categories of $\mathcal{D}_A$ are missing from $\mathcal{D}_B$ and can be tested in one direction only.

X. CONCLUSION AND REPORTING REQUIREMENTS

A held-out score on one corpus says little about how a detection xApp will behave in another deployment. On the corpora we examined, a random split inflates the score, and a model-free lookup gains about as much, which fits the recognition of capture sessions. The radio corpus's capture schedule rules out any estimate of temporal drift, and its false positive rate depends largely on which benign session a detector meets. The published 99.9% on the target corpus depends on record-position fields. Half of that corpus lies on records that occur under both labels, and the addresses of the benign copies identify them as flood traffic from a second attacker. On flows with consistent labels, transferred detectors reach a balanced accuracy of 0.61 to 0.78, below what the target corpus itself allows. On a third corpus they catch the SYN flood but flag much of the benign traffic. At a realistic base rate no threshold gives usable operational precision. Only latency passes. Through a FlexRIC loop with an emulated E2 node, a decision and its control message take at most 5.01 ms at the 99th percentile, most of it spent outside inference. We therefore propose that any study claiming a deployable O-RAN detection capability report at least the following:

- 1) performance on a corpus collected independently of the training data, in a prevalence-invariant metric such as balanced accuracy or ROC-AUC, beside input-blind baselines and a reference trained inside that corpus;
- 2) the split protocol and its group key, results under a group-disjoint split, and the category coverage of any time-based split;
- 3) the false positive rate on benign traffic with its unit, the resulting alert rate, and the operational precision at a stated base rate, all from pooled confusion counts with intervals that resample clusters;
- 4) the operating threshold, how it was chosen, and the recall it gives up;
- 5) intervals that account for overlap between resampled splits;
- 6) tail latency of the loop from indication to control acknowledgment, with its stages, implementation, reporting period, and host, measured through a RIC and, before any conformance claim, with a real E2 node;
- 7) evidence that the labels are a function of the features used, with no identical records under opposite labels, and that no record-position or identifier field is a feature.

REFERENCES

- [1] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 1376–1411, 2023. doi: 10.1109/COMST.2023.3239220
- [2] A. S. Abdalla, P. S. Upadhyaya, V. K. Shah, and V. Marojevic, "Toward next generation open radio access networks: What O-RAN can and cannot do!" *IEEE Network*, vol. 36, no. 6, pp. 206–213, 2022. doi: 10.1109/MNET.108.2100659
- [3] S. Niknam, A. Roy, H. S. Dhillon, S. Singh, R. Banerji, J. H. Reed, N. Saxena, and S. Yoon, "Intelligent O-RAN for beyond 5G and 6G wireless networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2022, pp. 215–220. doi: 10.1109/GCWkshps56602.2022.10008676
- [4] L. Bonati, S. D'Oro, M. Polese, S. Basagni, and T. Melodia, "Intelligence and learning in O-RAN for data-driven NextG cellular networks," *IEEE Communications Magazine*, vol. 59, no. 10, pp. 21–27, 2021. doi: 10.1109/MCOM.101.2001120
- [5] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis, D. Kumar, C. Lever, Z. Ma, J. Mason, D. Menscher, C. Seaman, N. Sullivan, K. Thomas, and Y. Zhou, "Understanding the Mirai botnet," in *Proc. 26th USENIX Security Symposium*, 2017, pp. 1093–1110. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis>
- [6] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "DDoS in the IoT: Mirai and other botnets," *Computer*, vol. 50, no. 7, pp. 80–84, 2017. doi: 10.1109/MC.2017.201
- [7] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "CoO-RAN: Developing machine learning-based xApps for open RAN closed-loop control on programmable experimental platforms," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 5787–5800, 2023. doi: 10.1109/TMC.2022.3188013
- [8] R. Schmidt, M. Irazabal, and N. Nikaein, "FlexRIC: An SDK for next-generation SD-RANs," in *Proc. ACM CoNEXT*, 2021, pp. 411–425. doi: 10.1145/3485983.3494870
- [9] H. Wen, P. Porras, V. Yegneswaran, A. Gehani, and Z. Lin, "5G-Spector: An O-RAN compliant layer-3 cellular attack detection service," in *Proc. Network and Distributed System Security Symposium (NDSS)*, 2024. doi: 10.14722/ndss.2024.24527
- [10] A. Scalingi, S. D'Oro, F. Restuccia, T. Melodia, and D. Giustiniano, "Det-RAN: Data-driven cross-layer real-time attack detection in 5G open RANs," in *Proc. IEEE Conf. on Computer Communications (INFOCOM)*, 2024, pp. 41–50. doi: 10.1109/INFOCOM52122.2024.10621223
- [11] S. Samarakoon, Y. Siriwardhana, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Ylianttila, "5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network," arXiv:2212.01298, 2022. [Online]. Available: <https://arxiv.org/abs/2212.01298>
- [12] L. Ilias, G. Doukas, V. Lamprou, C. Ntanos, and D. Askounis, "Convolutional neural networks and mixture of experts for intrusion detection in 5G networks and beyond," arXiv:2412.03483, 2024, the weighted F_1 cited is from the abstract of v2 (May 2025); a revised version appeared in *Front. Artif. Intell.*, vol. 8, art. 1708953, 2026, doi:10.3389/frai.2025.1708953. [Online]. Available: <https://arxiv.org/abs/2412.03483>
- [13] S. Layeghy and M. Portmann, "Explainable cross-domain evaluation of ML-based network intrusion detection systems," *Computers and Electrical Engineering*, vol. 108, p. 108692, 2023. doi: 10.1016/j.compeleceng.2023.108692
- [14] S. Layeghy, M. Gallagher, and M. Portmann, "Benchmarking the benchmark — comparing synthetic and real-world network IDS datasets," *Journal of Information Security and Applications*, vol. 80, p. 103689, 2024. doi: 10.1016/j.jisa.2023.103689
- [15] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security*, vol. 3, no. 3, pp. 186–205, 2000. doi: 10.1145/357830.357849
- [16] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *Proc. IEEE Symposium on Security and Privacy*, 2010, pp. 305–316. doi: 10.1109/SP.2010.25
- [17] ETSI, "Publicly available specification (PAS); E2 interface: General aspects and principles (O-RAN.WG3.E2GAP-R003-v04.01)," European Telecommunications Standards Institute, ETSI TS 104 038 V4.1.0, Oct. 2024. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/104000_104099/104038/04.01.00_60/ts_104038v040100p.pdf
- [18] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *Proc. 31st USENIX Security Symposium*, 2022, pp. 3971–3988. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity22/presentation/arp>

- [19] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in *Proc. 28th USENIX Security Symposium*, 2019, pp. 729–746.
- [20] A. Abrahem and A. Edhirig, "Bidirectional cross-dataset generalization and label-efficient adaptation for 5G network intrusion detection," *Wadi Alshatti University Journal of Pure and Applied Sciences*, vol. 4, no. 2, Aug. 2026. [Online]. Available: <https://waujpas.com/index.php/journal/article/view/509>
- [21] L. Obiuewvi, K. J. Rechowicz, S. Jayarathna, S. H. Bouk, F. Afrin, C. N. Barati, N. Moghim, V. Nannou, M. E. Ur Rahman, and S. Shetty, "Enabling real-time AI in O-RAN: Deploying and measuring AI inside a near-RT RIC xApp," in *Proc. 35th Int. Conf. on Computer Communications and Networks (ICCCN)*, Jul. 2026, pp. 1–9. doi: 10.1109/ICCCN69946.2026.11662852
- [22] B. Nugraha, M. Hajizadeh, T. Niehoff, A. V. Jnanashree, T. V. Phan, D. Triantafyllopoulou, O. Krause, M. Mieth, K. Moessner, and T. Bauschert, "A comprehensive 5G dataset for control and data plane security and resource management," in *Proc. IEEE International Conference on Cyber Security and Resilience (CSR)*, 2025, pp. 326–333. doi: 10.1109/CSR64739.2025.11130023
- [23] L. Bonati, M. Polese, S. D'Oro, S. Basagni, and T. Melodia, "OpenRAN Gym: AI/ML development, data collection, and testing for O-RAN on PAWR platforms," *Computer Networks*, vol. 220, p. 109502, 2023. doi: 10.1016/j.comnet.2022.109502
- [24] L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "SCOPE: An open and softwareized prototyping platform for NextG systems," in *Proc. ACM MobiSys*, 2021, pp. 415–426. doi: 10.1145/3458864.3466863
- [25] N. Nikaein, M. K. Marina, S. Manickam, A. Dawson, R. Knopp, and C. Bonnet, "OpenAirInterface: A flexible platform for 5G research," *ACM SIGCOMM Computer Communication Review*, vol. 44, no. 5, pp. 33–38, 2014. doi: 10.1145/2677046.2677053
- [26] I. Gomez-Miguel, A. Garcia-Saavedra, P. D. Sutton, P. Serrano, C. Cano, and D. J. Leith, "srsLTE: An open-source platform for LTE evolution and experimentation," in *Proc. ACM WINTech*, 2016, pp. 25–32. doi: 10.1145/2980159.2980163
- [27] ETSI, "Publicly available specification (PAS); O-RAN security threat modeling and risk assessment (O-RAN.WG11.Threat-Modeling.O-R003-v03.00)," European Telecommunications Standards Institute, ETSI TR 104 106 V3.0.0, Jun. 2025. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/104100_104199/104106/03.00.00_60/tr_104106v030000p.pdf
- [28] M. Liyanage, A. Braeken, S. Shahabuddin, and P. Ranaweera, "Open RAN security: Challenges and opportunities," *Journal of Network and Computer Applications*, vol. 214, p. 103621, 2023. doi: 10.1016/j.jnca.2023.103621
- [29] D. Mimran, R. Bitton, Y. Kfir, E. Klevansky, O. Brodt, H. Lehmann, Y. Elovici, and A. Shabtai, "Security of open radio access networks," *Computers & Security*, vol. 122, p. 102890, 2022. doi: 10.1016/j.cose.2022.102890
- [30] J. Groen, S. D'Oro, U. Demir, L. Bonati, M. Polese, T. Melodia, and K. R. Chowdhury, "Implementing and evaluating security in O-RAN: Interfaces, intelligence, and platforms," *IEEE Network*, vol. 39, no. 1, pp. 227–234, 2025. doi: 10.1109/MNET.2024.3434419
- [31] H. Fard, I. Komarov, and G. Wunder, "Cross-layer intrusion detection in 5G O-RAN: Gains and limits of fusing radio telemetry with network flow records," arXiv:2606.22450, 2026. [Online]. Available: <https://arxiv.org/abs/2606.22450>
- [32] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset," *Future Generation Computer Systems*, vol. 100, pp. 779–796, 2019. doi: 10.1016/j.future.2019.05.041
- [33] N. Moustafa, "A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets," *Sustainable Cities and Society*, vol. 72, p. 102994, 2021. doi: 10.1016/j.scs.2021.102994
- [34] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning," *IEEE Access*, vol. 10, pp. 40 281–40 306, 2022. doi: 10.1109/ACCESS.2022.3165809
- [35] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, 2023. doi: 10.3390/s23135941
- [36] I. Sharafaldin, A. Habibi Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. on Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116. doi: 10.5220/0006639801080116
- [37] S. Samarakoon, Y. Siriwardhana, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Ylianttila, "5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network," *IEEE DataPort*, 2022. doi: 10.21227/xtep-hv36
- [38] Y. Siriwardhana, S. Samarakoon, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Ylianttila, "Descriptor: 5G wireless network intrusion detection dataset (5G-NIDD)," *IEEE Data Descriptions*, vol. 2, pp. 358–369, 2025. doi: 10.1109/IEEEDATA.2025.3592888
- [39] F. A. Zadeh, A. Civciss, V. Ravihansa, C. Sandeepa, and M. Liyanage, "Descriptor: 5G open radio access network multi-modal intrusion detection dataset (NetsLab-5GORAN-IDD)," *IEEE Data Descriptions*, vol. 2, pp. 348–357, 2025. doi: 10.1109/IEEEDATA.2025.3614167
- [40] M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, "NetFlow datasets for machine learning-based network intrusion detection systems," in *Big Data Technologies and Applications (BDTA/WiCon)*, LNICST, vol. 371, 2021, pp. 117–135. doi: 10.1007/978-3-030-72802-1_9
- [41] M. Sarhan, S. Layeghy, and M. Portmann, "Towards a standard feature set for network intrusion detection system datasets," *Mobile Networks and Applications*, vol. 27, no. 1, pp. 357–370, 2022. doi: 10.1007/s11036-021-01843-0
- [42] G. Engelen, V. Rimmer, and W. Joosen, "Troubleshooting an intrusion detection dataset: the CICIDS2017 case study," in *Proc. IEEE Security and Privacy Workshops (SPW)*, 2021, pp. 7–12. doi: 10.1109/SPW53761.2021.00009
- [43] R. Flood, G. Engelen, D. Aspinall, and L. Desmet, "Bad design smells in benchmark NIDS datasets," in *Proc. IEEE European Symposium on Security and Privacy (EuroS&P)*, 2024, pp. 658–675. doi: 10.1109/EuroSP60621.2024.00042
- [44] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023. doi: 10.1016/j.patter.2023.100804
- [45] G. Apruzzese, P. Laskov, E. Montes de Oca, W. Mallouli, L. Brdalo Rapa, A. V. Grammatopoulos, and F. Di Franco, "The role of machine learning in cybersecurity," *Digital Threats: Research and Practice*, vol. 4, no. 1, pp. 1–38, 2023. doi: 10.1145/3545574
- [46] D. Kus, E. Wagner, J. Pennenkamp, K. Wolsing, I. B. Fink, M. Dahlmans, K. Wehrle, and M. Henze, "A false sense of security? Revisiting the state of machine learning-based industrial intrusion detection," in *Proc. ACM Cyber-Physical System Security Workshop (CPSS)*, 2022, pp. 73–84. doi: 10.1145/3494107.3522773
- [47] L. D'hooge, T. Wauters, B. Volckaert, and F. De Turck, "Inter-dataset generalization strength of supervised machine learning methods for intrusion detection," *Journal of Information Security and Applications*, vol. 54, p. 102564, 2020. doi: 10.1016/j.jisa.2020.102564
- [48] M. Saerens, P. Latinne, and C. Decaestecker, "Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure," *Neural Computation*, vol. 14, no. 1, pp. 21–41, 2002. doi: 10.1162/089976602753284446
- [49] Z. C. Lipton, Y.-X. Wang, and A. J. Smola, "Detecting and correcting for label shift with black box predictors," in *Proc. 35th Int. Conf. on Machine Learning (ICML)*, 2018, pp. 3122–3130.
- [50] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [51] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [52] T. Fawcett and A. Niculescu-Mizil, "PAV and the ROC convex hull," *Machine Learning*, vol. 68, no. 1, pp. 97–106, 2007. doi: 10.1007/s10994-007-5011-0
- [53] S. Rabanser, S. Günnemann, and Z. C. Lipton, "Failing loudly: An empirical study of methods for detecting dataset shift," in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [54] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proc. 30th AAAI Conf. on Artificial Intelligence*, 2016, pp. 2058–2065. doi: 10.1609/aaai.v30i1.10306
- [55] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003. doi: 10.1023/A:1024068626366
- [56] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. doi: 10.1023/A:1010933404324

- [57] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. doi: 10.1145/2939672.2939785
- [58] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://jmlr.org/papers/v12/pedregosa11a.html>
- [59] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734. doi: 10.3115/v1/D14-1179
- [60] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. on Data Mining (ICDM)*, 2008, pp. 413–422. doi: 10.1109/ICDM.2008.17
- [61] X. Jin, L. E. Li, L. Vanbever, and J. Rexford, "SoftCell: Scalable and flexible cellular core network architecture," in *Proc. 9th ACM Conf. on Emerging Networking Experiments and Technologies (CoNEXT)*, 2013, pp. 163–174. doi: 10.1145/2535372.2535377
- [62] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017, pp. 4765–4774. [Online]. Available: <https://papers.nips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>

ARSHAD ALI is with the Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia.

ADEEL AHMAD is with the Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia.

• • •